Bicomplex Difference Calculus on Native Binary Rings

Boolean Möbius Conjugacy, Recursive Center Nesting, and Reader-Relative Realization

Driven by Dean A. Kulik

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Abstract

Constrained discrete dynamical systems whose global trajectories are generated by local involutions exhibit a structural tension between state-value representation and relational transport. In this paper, we construct a unified operator calculus on the state-space matrix $X \in \mathbb{F}_2^{2^n \times n}$ of the native binary ring R_n, establishing an exact algebraic bridge between circular successor transport, word-internal channel decomposition, Boolean Möbius transduction, recursive center nesting, and physical realization.

First, we formulate the **Two-Axis Bicomplex Theorem**, proving that left-multiplication on X governs the address-transport axis ($\mathbb{F}_2^{2^n}$) while right-multiplication governs the word-internal bit axis ($\mathbb{F}_2^n$), so that every address-space operator commutes identically with the word-internal shape projection G_δ and shell-lamp projection G_{lamp}. Second, using Lucas's theorem in the group algebra $\mathbb{F}_2[T]/(T^{2^n} - I)$, we prove the **Boolean Möbius Conjugacy Identity** $MD = SA$, where M is the involutive lower-triangular Boolean Möbius transform ($M^2 = I$), $A = MX$ is the spectral address representation, $D = (I + T)X$ is the cyclic adjacent-state XOR instruction stream, and $S = M(I + T)M$ is the nilpotent superdiagonal shift. Because the native ring originates at $R_n[0] = 0^n$, the zero-frequency Möbius mode vanishes ($A_{0,\bullet} = 0^n$), yielding the lossless state reconstruction identity $X = MS^{\top}MD$. Third, we derive the matrix-level intertwining law $\Pi_{\text{step}}D = DJ_n \oplus \phi(1^n)^{\top}$, demonstrating that the instruction stream is an exact self-mirror in the shape channel ($\Pi_{\text{step}}\Delta(D) = \Delta(D)J_{n-1}$) while isolating all palindrome boundary defects $\phi = (I + T)p$ inside the rank-one face channel. Fourth, we establish **Palindrome Isolation** across all widths $n \geq 2$—showing that no two palindromes are adjacent in R_n and fixing the palindrome flag count at $2^{\lceil n/2 \rceil + 1} - 2$—and supply the complete inductive all-width proof of the quadratic closed-form instruction alphabet $\mathcal{A}_n$ ($|\mathcal{A}_n| = n^2/2 + 4$ for even $n \geq 6$ and $(n^2 + 3)/2$ for odd $n \geq 6$), resolving an open proof obligation in the prior record.

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Finally, we connect the ring's operator algebra to the Ho–Kalman realization factorization $M_{\mathrm{rel}} = FV^{\top}$ on finite symmetry arenas and evaluate its physical compilation in protein β-hairpins across a 160-chain, 143-hairpin *Escherichia coli* structural benchmark. While the shape-channel zipper word $Z = \tau \cdot \Lambda_2 \cdot \Lambda_4 \cdots$ recovers 887 of 1,062 directed backbone hydrogen bonds and the pre-sealed sequence-only face-coherence operator $Q(S, w)$ separates true creases from composition shuffles, parity-matched shifts, and detector-negative decoys (mean rank advantage 0.634060 vs. 0.498424, 0.513876, and 0.475469), unique sequence-only crease localization (12 of 132 ranked first) and physical sheet-plane orientation require nonlocal cross-arm registration. A constructive Euclidean counterexample demonstrates that a single covalent N–C_α–C–C_β tetrahedron cannot identify the β-sheet normal, and the 41 un-emitted $S \pm 1$ hydrogen-bond slips map to the conjugate dihedral cell reflections TP and PT.

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1. Introduction: The Executable Representation Problem

In a constrained discrete system whose state space is finite and whose evolution is governed by fixed-point-free involutions, global trajectories are determined by local state-to-state differences. A central mathematical problem in theoretical computation and mathematical physics is to determine how a global cyclic program is encoded within local differential instructions, which structural invariants are preserved under coordinate transformations, and what additional geometric or boundary information is required to compile an abstract instruction stream into a realized physical structure.

When state spaces are indexed by n-bit binary words $x \in \mathbb{B}_n = \{0,1\}^n$ of size $N = 2^n$, two distinct notions of difference arise that must be distinguished:

1. **Word-internal difference (the shape axis):** Given a single state word $x = (b_0, b_1, \dots, b_{n-1})$, its internal adjacency structure is captured by the $(n-1)$-bit differential word $\delta_i(x) = b_i \oplus b_{i+1}$, which together with the leading face bit $f(x) = b_0$ forms an invertible coordinate split (f, δ).
2. **Address-step difference (the instruction axis):** Given a cyclic ordering of all N states into a ring $R_n = (R_n[0], R_n[1], \dots, R_n[N-1])$, the step-to-step transition is carried by the adjacent-state XOR instruction stream $D_k = R_n[k] \oplus R_n[(k+1) \bmod N]$.

Establishing how these two differential structures interact—and how both transform under the Boolean Möbius transducer M over $\mathbb{F}_2$—requires a two-axis matrix formalism on the full ring state table $X \in \mathbb{F}_2^{N \times n}$. Once the ring is represented as an $(N \times n)$ matrix over $\mathbb{F}_2$, operations that re-index or transport addresses act from the left, while operations that project or permute bit positions within each word act from the right.

Developing this bicomplex difference calculus yields four principal results across algebra, combinatorics, representation-theoretic realization, and macromolecular geometry. First, in Section 3.1, we show that address transport T, address reflection Π_N, cyclic differentiation $I_N \oplus T$, and the Boolean Möbius transform M commute identically with the word-internal shape operator G_δ and shell-lamp operator G_{lamp}. Global reflection in the native ring consequently acts as an exact linear involution on the shape channel while concentrating all non-linear palindrome-branching defects into a rank-one perturbation of the face channel. Second, in Section 3.2, we prove that the involutive Boolean Möbius matrix M conjugates the cyclic difference operator $I_N \oplus T$ into the nilpotent superdiagonal shift $S = M(I_N \oplus T)M$ over $\mathbb{F}_2$, yielding the exact operator identity $MD = SA$ where $A = MX$. Because the native ring starts at 0^n, this relation is losslessly invertible via $X = MS^{\top}MD$.

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Third, in Section 3.3 and Section 4, whereas the closed-form instruction alphabet $\mathcal{A}_n$ of the native ring was previously established computationally through $n = 16$ (with widths 13 through 16 verified against a pre-execution SHA-256 seal) while its all-width proof remained open, we supply the complete inductive proof for all $n \geq 6$. We also prove that palindromes are strictly isolated in the ring walk for every $n \geq 2$, locking the number of face-channel correction flags at $2^{\lceil n/2 \rceil + 1} - 2$. Fourth, in Section 5 and Section 6, we show that the separation between an underlying relational program and its observable projection is governed by a Ho–Kalman input-output factorization $M_{\mathrm{rel}} = FV^{\top}$ over rational irreducible representations. Applying this distinction to the compilation of protein β-hairpins on the residue-pair square $W_m(i,j) = i \parallel J(j)$, we demonstrate what the ring's local operators generate (the antiparallel zipper $Z = \tau \cdot \Lambda_2 \cdot \Lambda_4 \cdots$, structure-conditioned face burial, and statistically separated sequence-only face-coherence ranking) and prove via constructive counterexamples what they cannot supply without an independent nonlocal registration input (unique crease address selection, local tetrahedral identification of the β-sheet plane, and the 30 to 11 empirical asymmetry between the conjugate dihedral slip reflections TP and PT).

2. Foundations: The Dihedral Involution Algebra and the Native Binary Ring

2.1 Binary Words, Operators J, C, P, and the Frame Law

Let $\mathbb{B}_n = \{0,1\}^n$ denote the set of n-bit binary words for $n \geq 2$, indexed from the most significant bit b_0 to the least significant bit b_{n-1}. We identify each word $x = (b_0, b_1, \ldots, b_{n-1}) \in \mathbb{B}_n$ with its standard unsigned integer value:

$$v(x) = \sum_{k=0}^{n-1} b_k \, 2^{n-1-k} \in \{0,1,\ldots,N-1\}, \qquad N = 2^n.$$

Under bitwise exclusive-OR ($\oplus$), $(\mathbb{B}_n, \oplus)$ is an n-dimensional vector space over the finite field $\mathbb{F}_2$. We define three fundamental involutions on $\mathbb{B}_n$:

Definition 2.1 (Reversal, Complement, and Native Partner). For any $x = (b_0, b_1, \ldots, b_{n-1}) \in \mathbb{B}_n$:

1. **Bit-reversal (J):** $J(b_0, b_1, \ldots, b_{n-1}) = (b_{n-1}, \ldots, b_1, b_0)$.
2. **Bitwise complement (C):** $C(x) = x \oplus 1^n = (1 \oplus b_0, 1 \oplus b_1, \ldots, 1 \oplus b_{n-1})$, or in integer value $C(x) = (N-1) - x$.
3. **Palindrome set (Pal_n):** $\mathrm{Pal}_n = \{x \in \mathbb{B}_n : J(x) = x\}$, which is a linear subspace of $\mathbb{B}_n$ over $\mathbb{F}_2$ of dimension $\lceil n/2 \rceil$ and cardinality $|\mathrm{Pal}_n| = 2^{\lceil n/2 \rceil}$.
4. **Native partner (P):** The piecewise involution

$$P(x) = \begin{cases} C(x) & \text{if } x \in \mathrm{Pal}_n, \\ J(x) & \text{if } x \notin \mathrm{Pal}_n. \end{cases}$$

Proposition 2.2 (Fixed-Point-Free Partner Involution). *For every $n \geq 2$, the operators J and C commute ($JC = CJ$), C preserves Pal_n, and $P: \mathbb{B}_n \to \mathbb{B}_n$ is a fixed-point-free involution ($P^2 = \mathrm{id}$, $P(x) \neq x$ for all $x \in \mathbb{B}_n$) that partitions $\mathbb{B}_n$ into $h = 2^{n-1}$ disjoint 2-element orbits.*

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Proof. Since $(1 \oplus b_0, \dots, 1 \oplus b_{n-1})$ reversed is $(1 \oplus b_{n-1}, \dots, 1 \oplus b_0)$, $J(C(x)) = C(J(x))$. Thus $J(x) = x \Leftrightarrow J(C(x)) = C(x)$, meaning $x \in \mathrm{Pal}_n \Leftrightarrow C(x) \in \mathrm{Pal}_n$. Consequently, if $x \in \mathrm{Pal}_n$, then $P(x) = C(x) \in \mathrm{Pal}_n$, so $P(P(x)) = C(C(x)) = x$, and $C(x) = x \oplus 1^n \neq x$. If $x \notin \mathrm{Pal}_n$, then $J(x) \neq x$ and $J(x) \notin \mathrm{Pal}_n$, so $P(P(x)) = J(J(x)) = x$. Hence $P^2 = \mathrm{id}$ and P has no fixed points. ■

Each 2-element orbit $\{x, P(x)\}$ has a unique smaller element under the standard integer order, which we call a **representative**:

$$\mathrm{Rep}_n = \{x \in \mathbb{B}_n : x < P(x)\}, \qquad |\mathrm{Rep}_n| = h = 2^{n-1}.$$

Every word $x \in \mathbb{B}_n$ for $n \geq 3$ can be written by peeling its outer shell $(b_0, b_{n-1}) \in \mathbb{B}_2$ around an $(n-2)$-bit core $y \in \mathbb{B}_{n-2}$ as $x = b_0 \cdot y \cdot b_{n-1}$. The classification of representatives by their outer shell is governed by the **Frame Law** (established in the *NESTED* record and verified in §195–§196):

Lemma 2.3 (The Frame Law). *For $n \geq 3$, a word $x = b \cdot y \cdot c$ with $b, c \in \{0,1\}$ and $y \in \mathbb{B}_{n-2}$ belongs to Rep_n if and only if one of the following three mutually exclusive conditions holds*:

1. $(b, c) = (0,1)$ *and* $y \in \mathbb{B}_{n-2}$ *is arbitrary (2^{n-2} words);*
2. $(b, c) = (0,0)$ *and* $y \leq J(y)$ *in* $\mathbb{B}_{n-2}$ *($2^{n-3} + 2^{\lceil (n-2)/2 \rceil - 1}$ words);*
3. $(b, c) = (1,1)$ *and* $y < J(y)$ *in* $\mathbb{B}_{n-2}$ *($\ell(n) = 2^{n-3} - 2^{\lfloor (n-3)/2 \rfloor}$ words for $n \geq 4$, and 0 words for $n = 3$). No representative ever has outer shell $(b, c) = (1,0)$.*

Proof. When $(b, c) = (0,1)$, $b \neq c$ implies $x \notin \mathrm{Pal}_n$ and $P(x) = J(x) = 1 \cdot J(y) \cdot 0$; since x starts with 0 and $P(x)$ starts with 1, $x < P(x)$ holds for every $y \in \mathbb{B}_{n-2}$. Conversely, when $(b, c) = (1,0)$, $P(x) = J(x) = 0 \cdot J(y) \cdot 1 < x$, so $x \notin \mathrm{Rep}_n$. When $(b, c) = (0,0)$, either $y = J(y)$ (so $x \in \mathrm{Pal}_n$ and $P(x) = C(x) = 1 \cdot C(y) \cdot 1 > x$), or $y \neq J(y)$ (so $x \notin \mathrm{Pal}_n$ and $P(x) = 0 \cdot J(y) \cdot 0 > x \Leftrightarrow y < J(y)$), which combine into $y \leq J(y)$. Finally, when $(b, c) = (1,1)$, if $y = J(y)$ then $x \in \mathrm{Pal}_n$ and $P(x) = C(x) = 0 \cdot C(y) \cdot 0 < x$, so $x \notin \mathrm{Rep}_n$; if $y \neq J(y)$, then $P(x) = J(x) = 1 \cdot J(y) \cdot 1 > x \Leftrightarrow y < J(y)$. ■

2.2 Construction of the Native Ring R_n and the Facing Law

Sorting Rep_n in strictly ascending integer order yields the list $r_0 < r_1 < \cdots < r_{h-1}$, where $r_0 = 0^n$ and $m(n) = r_{h-1}$ is the maximum representative.

Definition 2.4 (The Native Binary Ring R_n). For $n \geq 2$, the native binary ring $R_n = (R_n[0], R_n[1], \dots, R_n[N-1])$ is the cyclic sequence of all $N = 2^n$ binary words constructed by listing the first $h - 1$ representatives in ascending order, swapping the seam pair $(P(m(n)), m(n))$ at the center indices $(h-1, h)$, and appending the partner images of the first $h - 1$ representatives in descending order:

$$R_n = (r_0, r_1, \dots, r_{h-2},\ P(m(n)),\ m(n),\ P(r_{h-2}), \dots, P(r_1), P(r_0)).$$

Let $\mathrm{pos}_n : \mathbb{B}_n \to \mathbb{Z}_N$ denote the inverse address lookup $\mathrm{pos}_n(R_n[k]) = k$, and let the **native successor** $T : \mathbb{B}_n \to \mathbb{B}_n$ be one cyclic step along the ring: $T(R_n[k]) = R_n[(k+1) \bmod N]$.

Proposition 2.5 (The Facing Law and Dihedral Conjugacy). *For every $n \geq 2$ and every address $k \in \{0,1, \dots, N-1\}$:*

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1. **Facing Law:** $R_n[N-1-k] = P(R_n[k])$, or in address coordinates $\mathrm{pos}_n(P(x)) + \mathrm{pos}_n(x) = N - 1$.
2. **Dihedral Inversion:** $P \circ T \circ P = T^{-1}$, so that $\langle T, P \rangle \cong D_N$ generates the dihedral group of order $2N = 2^{n+1}$ on $\mathbb{B}_n$.
3. **Palindromic Facing Masks:** For every $x \in \mathbb{B}_n$, the facing instruction $\Lambda(x) = x \oplus P(x)$ is a nonzero palindrome ($\Lambda(x) \in \mathrm{Pal}_n \backslash \{0^n\}$).

Proof. For $k \in \{0, \dots, h-2\}$, $R_n[k] = r_k$ and $R_n[N-1-k] = P(r_k) = P(R_n[k])$ by Definition 2.4. For $k = h-1$, $R_n[h-1] = P(m(n))$ and $R_n[h] = m(n) = P(P(m(n))) = P(R_n[h-1])$ since $P^2 = \mathrm{id}$. This proves (1) for all k. Consequently, in address coordinates $k \in \mathbb{Z}_N$, P acts as $k \mapsto N-1-k \equiv -1-k \pmod N$ while T acts as $k \mapsto k+1 \pmod N$. Thus $P(T(P(k))) \equiv -1-(-1-k+1) \equiv k-1 \pmod N = T^{-1}(k)$, proving (2). Finally, if $x \in \mathrm{Pal}_n$, $x \oplus P(x) = x \oplus C(x) = 1^n \in \mathrm{Pal}_n \backslash \{0^n\}$; if $x \notin \mathrm{Pal}_n$, $J(x \oplus J(x)) = J(x) \oplus x = x \oplus J(x) \neq 0^n$, proving (3). ■

2.3 The Invertible Face/Shape Decomposition and Shell Lamps

To track how the involutions J, C, P act on individual words, we decompose $\mathbb{B}_n$ into value (face) and differential (shape) coordinates (*From One Difference to the Shape Channel* §3.2; §195–§196).

Definition 2.6 (Face Channel, Shape Channel, and Shell Lamps). For any $x = (b_0, b_1, \dots, b_{n-1}) \in \mathbb{B}_n$:

1. **Face (value) channel:** $f: \mathbb{B}_n \to \mathbb{F}_2$ is the leading bit $f(x) = b_0$.
2. **Shape (difference) channel:** $\delta: \mathbb{B}_n \to \mathbb{F}_2^{n-1}$ is the adjacent-bit XOR vector
$$\delta(x) = (\delta_0(x), \delta_1(x), \dots, \delta_{n-2}(x)), \qquad \delta_i(x) = b_i \oplus b_{i+1}.$$
3. **Shell lamps:** $g: \mathbb{B}_n \to \mathbb{F}_2^{\lfloor n/2 \rfloor}$ is the nested concentric shell-difference vector
$$g(x) = (g_0(x), g_1(x), \dots, g_{\lfloor n/2 \rfloor - 1}(x)), \qquad g_k(x) = b_k \oplus b_{n-1-k}.$$

Proposition 2.7 (Channel Separation of J, C, P). *For all $n \geq 2$ and all $x \in \mathbb{B}_n$:*

1. *The map $x \mapsto (f(x), \delta(x))$ is an $\mathbb{F}_2$-linear bijection $\mathbb{B}_n \xrightarrow{\sim} \mathbb{F}_2 \times \mathbb{F}_2^{n-1}$ with inverse $b_0 = f$, $b_{i+1} = b_i \oplus \delta_i = f \oplus \bigoplus_{j=0}^{i} \delta_j$.*
2. *Complement C acts purely on the face channel: $f(C(x)) = 1 \oplus f(x)$ and $\delta(C(x)) = \delta(x)$.*
3. *Both J and P act on the shape channel by exact bit-reversal: $\delta(J(x)) = \delta(P(x)) = J_{n-1}(\delta(x))$.*
4. *The shell lamps $g(x)$ are invariant under all three operators: $g(J(x)) = g(C(x)) = g(P(x)) = g(x)$, and $x \in \mathrm{Pal}_n \Leftrightarrow g(x) = 0^{\lfloor n/2 \rfloor}$.*

Proof. Statement (1) holds by telescope summation over $\mathbb{F}_2$. Statement (2) follows because $(1 \oplus b_i) \oplus (1 \oplus b_{i+1}) = b_i \oplus b_{i+1}$. For (3), $\delta_i(J(x)) = b_{n-1-i} \oplus b_{n-2-i} = \delta_{n-2-i}(x) = (J_{n-1}(\delta(x)))_i$; since $P(x)$ is either $J(x)$ or $C(x) = C(J(x))$ (on palindromes), and C leaves δ invariant, $\delta(P(x)) = \delta(J(x)) = J_{n-1}(\delta(x))$ for every $x \in \mathbb{B}_n$. For (4), $b_{n-1-k} \oplus b_k = b_k \oplus b_{n-1-k}$ and $(1 \oplus b_k) \oplus (1 \oplus b_{n-1-k}) = b_k \oplus b_{n-1-k}$. ■

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3. The Two-Axis Bicomplex and Boolean Möbius Conjugacy

3.1 The State Matrix X and Commutation of Address and Word Axes

We now unify the address-indexed bit planes of R_n and the word-internal channels into a single matrix algebra over $\mathbb{F}_2$.

Definition 3.1 (Ring State Matrix and Bit-Plane Operators). Let $X \in \mathbb{F}_2^{N\times n}$ ($N = 2^n$) be the state matrix of the native binary ring R_n, whose k-th row ($0 \le k \le N-1$) is the binary word $R_n[k]$ and whose j-th column ($0 \le j \le n-1$) is the j-th address-indexed bit plane $X_{\bullet,j} \in \mathbb{F}_2^N$:

$$X_{k,j} = \text{bit } j \text{ of } R_n[k].$$

Along the vertical address axis ($\mathcal{L} \subset \mathbb{F}_2^{N\times N}$), left-multiplication on X defines four core operators: the cyclic shift $T \in \mathbb{F}_2^{N\times N}$ with $T_{k,m} = \mathbf{1}[m \equiv k+1 \pmod N]$ so that $(TX)_{k,\bullet} = X_{(k+1) \bmod N,\bullet}$; the cyclic XOR difference operator $D_{\mathrm{op}} = I_N \oplus T \in \mathbb{F}_2^{N\times N}$, which generates the **instruction stream matrix** $D = (I_N \oplus T)X \in \mathbb{F}_2^{N\times n}$ with rows $D_{k,\bullet} = R_n[k] \oplus R_n[(k+1) \bmod N]$; the address reflection matrix $\Pi_N \in \mathbb{F}_2^{N\times N}$ with $(\Pi_N)_{k,m} = \mathbf{1}[m = N-1-k]$; and the step-address reflection matrix $\Pi_{\mathrm{step}} = T\Pi_N \in \mathbb{F}_2^{N\times N}$ with $(\Pi_{\mathrm{step}})_{k,m} = \mathbf{1}[m \equiv N-2-k \pmod N]$, which reflects the $N-1$ interior steps $k \leftrightarrow N-2-k$ and fixes the wrap step $k = N-1$.

Along the horizontal word axis ($\mathcal{R}$), right-multiplication on X defines the word bit-reversal permutation $J_n \in \mathbb{F}_2^{n\times n}$ with $(J_n)_{i,j} = \mathbf{1}[j = n-1-i]$; the face-bit column vector $e_0 = (1,0,\dots,0)^{\top} \in \mathbb{F}_2^{n\times 1}$ giving $f(X) = Xe_0 \in \mathbb{F}_2^{N\times 1}$; the shape generator matrix $G_\delta \in \mathbb{F}_2^{n\times(n-1)}$ with $(G_\delta)_{i,j} = \mathbf{1}[i=j] \oplus \mathbf{1}[i=j+1]$ giving $\Delta(X) = XG_\delta \in \mathbb{F}_2^{N\times(n-1)}$; and the shell-lamp generator matrix $G_{\mathrm{lamp}} \in \mathbb{F}_2^{n\times\lfloor n/2\rfloor}$ with $(G_{\mathrm{lamp}})_{i,j} = \mathbf{1}[i=j] \oplus \mathbf{1}[i=n-1-j]$ giving $L(X) = XG_{\mathrm{lamp}} \in \mathbb{F}_2^{N\times\lfloor n/2\rfloor}$.

Theorem 3.2 (Two-Axis Commutation and Bicomplex Exactness). For every address operator $L \in \mathbb{F}_2^{N\times N}$ and every word-internal linear operator $R \in \mathbb{F}_2^{n\times m}$, the left action of L and the right action of R commute on $\mathbb{F}_2^{N\times n}$: $(LX)R = L(XR)$. Furthermore, the horizontal word-difference sequence and vertical address-difference sequence form a commuting bicomplex over $\mathbb{F}_2$:

1. **Horizontal exactness (Word axis):** $\ker(G_\delta^{\top}: \mathbb{F}_2^n \to \mathbb{F}_2^{n-1}) = \mathrm{span}_{\mathbb{F}_2}\{1^n\}$, so $(1^n)^{\top}G_\delta = 0^{n-1}$.
2. **Vertical exactness (Address axis):** $\ker(I_N \oplus T: \mathbb{F}_2^N \to \mathbb{F}_2^N) = \mathrm{span}_{\mathbb{F}_2}\{1^N\}$ and $\mathrm{im}(I_N \oplus T) = \{v \in \mathbb{F}_2^N : \bigoplus_{k=0}^{N-1} v_k = 0\}$.
3. **Shape of the Instruction Stream:** $\Delta(D) = DG_\delta = ((I_N \oplus T)X)G_\delta = (I_N \oplus T)(XG_\delta) = (I_N \oplus T)\Delta(X)$.

Proof. Matrix multiplication over $\mathbb{F}_2$ is associative, so $(LX)R = L(XR)$ for all L, X, R. For (1), a row vector $w \in \mathbb{F}_2^n$ satisfies $wG_\delta = 0^{n-1}$ iff $w_i \oplus w_{i+1} = 0$ for all $0 \le i \le n-2$, iff $w_0 = w_1 = \cdots = w_{n-1}$, which spans $\{0^n, 1^n\}$. For (2), $(I_N \oplus T)v = 0^N$ iff $v_k = v_{(k+1) \bmod N}$ for all k, spanning $\{0^N, 1^N\}$, and since $\mathrm{rank}(I_N \oplus T) = N-1$ with $(1^N)^{\top}(I_N \oplus T) = 0^N$, the image is the even-weight hyperplane. Statement (3) is immediate from associativity. ■

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3.2 Boolean Möbius Transduction and the Identity $MD = SA$

We now examine the action of the Boolean Möbius transformation M (also known in coding theory and Boolean analysis as the Reed–Muller or Zhegalkin transform) on the address-indexed bit planes of R_n and establish the exact identity $MD = SA$.

Definition 3.3 (Boolean Möbius Transducer M and Superdiagonal Shift S). On the address space $\{0,1,\dots,N-1\}$ with $N = 2^n$, write $v \preccurlyeq u$ if the binary expansion of v is a bitwise subset of the binary expansion of u (i.e., $(v \& u) = v$).

1. The **Boolean Möbius matrix** $M \in \mathbb{F}_2^{N\times N}$ is the lower-triangular binary matrix defined by:

$$M_{u,v} = \mathbf{1}[v \preccurlyeq u] = \binom{u}{v} \bmod 2, \qquad u, v \in \{0,1,\dots,N-1\},$$

where the congruence $\binom{u}{v} \equiv \mathbf{1}[v \preccurlyeq u] \pmod 2$ is Lucas's Theorem.

2. The **superdiagonal shift matrix** $S \in \mathbb{F}_2^{N\times N}$ is the nilpotent matrix ($S^N = 0$) defined by:

$$S_{u,v} = \mathbf{1}[v = u + 1], \qquad 0 \le u, v \le N-1,$$

so that $(Sa)_u = a_{u+1}$ for $0 \le u \le N-2$ and $(Sa)_{N-1} = 0$.

Theorem 3.4 (Boolean Möbius Conjugacy $MD = SA$ and Lossless State Reconstruction). *Let $N = 2^n$ ($n \ge 1$). Over $\mathbb{F}_2$:*

1. *The Boolean Möbius matrix is self-inverse: $M^2 = I_N$.*
2. *M conjugates the cyclic difference operator $I_N \oplus T$ into the nilpotent superdiagonal shift S:*

$$M(I_N \oplus T)M = S, \qquad \text{or equivalently} \qquad M(I_N \oplus T) = SM.$$

3. *For any address-indexed bit plane $x \in \mathbb{F}_2^N$ with Möbius transform $a = Mx \in \mathbb{F}_2^N$ and cyclic adjacent-state XOR stream $d = (I_N \oplus T)x \in \mathbb{F}_2^N$, and for the full ring state matrix $X \in \mathbb{F}_2^{N\times n}$ with $A = MX$ and $D = (I_N \oplus T)X$, the following exact identities hold:*

$$Md = Sa, \qquad MD = SA, \qquad M\Delta(D) = S\Delta(A).$$

4. *Because the native ring R_n satisfies $R_n[0] = 0^n$, the top row of $A = MX$ vanishes ($A_{0,\bullet} = X_{0,\bullet} = 0^n$). Consequently, $S^\top SA = A$, and the full state matrix X is reconstructed losslessly from the instruction stream D via:*

$$X = MS^\top MD.$$

Proof.

1. For any $u, w \in \{0,\dots,N-1\}$, $(M^2)_{u,w} = \bigoplus_{v=0}^{N-1} \mathbf{1}[w \preccurlyeq v \preccurlyeq u] \pmod 2$. The interval $\{v : w \preccurlyeq v \preccurlyeq u\}$ is empty if $w \not\preccurlyeq u$, and is a Boolean hypercube of size $2^{\mathrm{wt}(u)-\mathrm{w}\ (w)}$ if $w \preccurlyeq u$. Over $\mathbb{F}_2$, $2^{\mathrm{wt}(u)-\mathrm{wt}(w)} \equiv 1 \pmod 2$ if and only if $\mathrm{wt}(u) = \mathrm{wt}(w)$, i.e., $u = w$. Thus $M^2 = I_N$.

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2. Consider the left-shift operator T on $\mathbb{F}_2^N$, which satisfies $Te_v = e_{(v-1) \bmod N}$ on the standard basis vectors $e_0, \ldots, e_{N-1}$ of $\mathbb{F}_2^N$. Equivalently, let $P_{\text{shift}} = T^{-1} = T^\top$ be the forward cyclic shift $P_{\text{shift}} e_v = e_{(v+1) \bmod N}$. In the polynomial algebra $\mathbb{F}_2[P_{\text{shift}}]/(P_{\text{shift}}^N - I_N)$, $(I_N \oplus P_{\text{shift}})^N = I_N \oplus P_{\text{shift}}^N = 0$ because $N = 2^n$ is a power of 2. For each $u \in \{0,1, \ldots, N-1\}$, applying Lucas's Theorem to the binomial expansion gives:

$$(I_N \oplus P_{\text{shift}})^u e_0 = \oplus_{v=0}^{u} \left(\binom{u}{v} \bmod 2 \right) P_{\text{shift}}^v e_0 = \oplus_{v=0}^{N-1} M_{u,v} e_v = (M^\top e_u).$$

Taking transposes, the u-th row of M, denoted $m_u^\top = e_u^\top M$, satisfies $m_u^\top = e_0^\top (I_N \oplus T)^u$. Now multiply $m_u^\top$ on the right by $(I_N \oplus T)$:

$$e_u^\top M (I_N \oplus T) = e_0^\top (I_N \oplus T)^u (I_N \oplus T) = e_0^\top (I_N \oplus T)^{u+1}.$$

For $0 \le u \le N-2$, $e_0^\top (I_N \oplus T)^{u+1} = m_{u+1}^\top = e_{u+1}^\top M = e_u^\top SM$. For $u = N-1$, $(I_N \oplus T)^N = I_N \oplus T^N = I_N \oplus I_N = 0$ over $\mathbb{F}_2$, so $e_{N-1}^\top M (I_N \oplus T) = 0 = e_{N-1}^\top SM$. Since this holds for every row $u \in \{0, \ldots, N-1\}$, we have $M(I_N \oplus T) = SM$, and multiplying on the right by $M = M^{-1}$ yields $M(I_N \oplus T)M = S$.

3. Multiplying $M(I_N \oplus T) = SM$ on the right by x (or X, or $\Delta(X) = XG_\delta$) gives $Md = M(I_N \oplus T)x = SMx = Sa$, $MD = SA$, and $M\Delta(D) = S\Delta(A)$.

4. Since $S_{u,v} = \mathbf{1}[v = u+1]$, the product $S^\top S \in \mathbb{F}_2^{N \times N}$ is the diagonal projection matrix $\text{diag}(0,1,1, \ldots, 1)$ that annihilates row 0 and acts as the identity on rows $1, \ldots, N-1$. For row $u = 0$ of $A = MX$, since $M_{0,v} = \mathbf{1}[v \leqslant 0] = \mathbf{1}[v = 0]$, we have $A_{0,\bullet} = X_{0,\bullet} = R_n[0] = 0^n$. Therefore $(S^\top SA)_{0,\bullet} = 0^n = A_{0,\bullet}$ and $(S^\top SA)_{u,\bullet} = A_{u,\bullet}$ for all $u \ge 1$, so $S^\top SA = A$. Substituting $SA = MD$ and $A = MX$ yields $MX = S^\top MD$, and multiplying by M on the left gives $X = MS^\top MD$. ■

Theorem 3.4 exposes a remarkable structural property of the Boolean Möbius transducer: in the spectral domain $A = MX$, the cyclic differentiation operator $I_N \oplus T$ does not mix modes; it shifts the spectral rows upward by one index ($A_{u+1,\bullet} = (MD)_{u,\bullet}$ for $0 \le u \le N-2$) and annihilates the constant mode $A_{0,\bullet}$. Simultaneously, the bottom row of MD vanishes identically ($(MD)_{N-1,\bullet} = (SA)_{N-1,\bullet} = 0^n$), reflecting the global closure constraint $\oplus_{k=0}^{N-1} D_{k,\bullet} = 0^n$ of the cyclic instruction stream.

3.3 Matrix Intertwining of the Instruction Stream and Palindrome Isolation

We next derive how the facing law $R_n[N-1-k] = P(R_n[k])$ intertwines the address-reflection operators Π_N and $\Pi_{\text{step}} = T\Pi_N$ with the word-reversal matrix J_n, and prove that the palindrome boundary flags are strictly isolated across all ring widths.

Theorem 3.5 (Matrix Intertwining and Channel Split of the Step Program). Let $p \in \mathbb{F}_2^{N \times 1}$ be the palindrome indicator column vector with $p_k = \mathbf{1}[R_n[k] \in \text{Pal}_n]$, and let $\phi = (I_N \oplus T)p \in \mathbb{F}_2^{N \times 1}$ be the palindrome boundary-crossing flag vector with $\phi_k = p_k \oplus p_{(k+1) \bmod N}$. Then for all $n \ge 2$:

1. **State Matrix Reflection:**

$$\Pi_N X = XJ_n \oplus p(1^n)^\top.$$

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2. **Instruction Stream Reflection:** The step-reversal operator $\Pi_{\text{step}} = T\Pi_N$ satisfies $\Pi_{\text{step}}(I_N \oplus T) = (I_N \oplus T)\Pi_N$, and therefore:

$$\Pi_{\text{step}} D = DJ_n \oplus \phi(1^n)^\top.$$

In row coordinates, for every $a \in \{0,1,\dots,N-2\}$, $D_{N-2-a} = J(D_a) \oplus \phi_a 1^n$, and at the wrap step $a = N-1$, $D_{N-1} = 1^n$ with $\phi_{N-1} = 0$.

3. **Exact Shape Mirror and Face Defect Isolation:** Right-multiplying (1) and (2) by the shape generator G_δ and face unit vector e_0 yields:

$$\Pi_N \Delta(X) = \Delta(X) J_{n-1}, \qquad \Pi_{\text{step}} \Delta(D) = \Delta(D) J_{n-1},$$

$$\Pi_{\text{step}} f(D) = De_{n-1} \oplus \phi.$$

4. **Möbius Conjugation of Reflection and Flags:** In the Boolean Möbius basis, the conjugated address-reflection operator $M\Pi_N M$ is the block-Toeplitz upper-triangular matrix $(M\Pi_N M)_{u,v} = \mathbf{1}[u \preccurlyeq v]$ (the transpose $M^\top$), and the Möbius transform of the flag vector satisfies $M\phi = S(Mp)$.

Proof.

1. By Definition 2.1, for any word $x \in \mathbb{B}_n$, $P(x) = J(x) \oplus \mathbf{1}[x \in \text{Pal}_n] \cdot 1^n$ because $J(x) = x$ and $C(x) = x \oplus 1^n$ on Pal_n. Applying the Facing Law $R_n[N-1-k] = P(R_n[k])$ to row k of X gives $(\Pi_N X)_{k,\bullet} = X_{k,\bullet} J_n \oplus p_k(1^n)^\top$, which in matrix form is $\Pi_N X = XJ_n \oplus p(1^n)^\top$.

2. Since $\Pi_N T \Pi_N = T^{-1}$, we have $T\Pi_N T = \Pi_N$. Therefore:

$$\Pi_{\text{step}}(I_N \oplus T) = T\Pi_N(I_N \oplus T) = T\Pi_N \oplus T\Pi_N T = T\Pi_N \oplus \Pi_N = (I_N \oplus T)\Pi_N.$$

Applying this to X gives $\Pi_{\text{step}} D = \Pi_{\text{step}}(I_N \oplus T)X = (I_N \oplus T)\Pi_N X = (I_N \oplus T)(XJ_n \oplus p(1^n)^\top) = DJ_n \oplus \phi(1^n)^\top$. At $a = N-1$, $R_n[N-1] = P(0^n) = 1^n \in \text{Pal}_n$ and $R_n[0] = 0^n \in \text{Pal}_n$, so $D_{N-1} = 1^n \oplus 0^n = 1^n$ and $\phi_{N-1} = 1 \oplus 1 = 0$.

3. By Theorem 3.2(1), $(1^n)^\top G_\delta = 0^{n-1}$. Furthermore, $(J_n G_\delta)_{i,j} = (G_\delta)_{n-1-i,j} = \mathbf{1}[n-1-i=j] \oplus \mathbf{1}[n-1-i=j+1] = \mathbf{1}[i=n-1-j] \oplus \mathbf{1}[i=n-2-j] = (G_\delta J_{n-1})_{i,j}$. Right-multiplying (1) and (2) by G_δ eliminates the rank-1 defect $p(1^n)^\top G_\delta = 0$ and $\phi(1^n)^\top G_\delta = 0$, leaving $\Pi_N \Delta(X) = \Delta(X) J_{n-1}$ and $\Pi_{\text{step}} \Delta(D) = \Delta(D) J_{n-1}$. Right-multiplying (2) by e_0 uses $J_n e_0 = e_{n-1}$ and $(1^n)^\top e_0 = 1$.

4. Applying Theorem 3.4(3) to the vector $p \in \mathbb{F}_2^N$ and $\phi = (I_N \oplus T)p$ gives $M\phi = S(Mp)$. For $M\Pi_N M$, note that Π_N maps index $v \mapsto (N-1) - v = v \oplus (N-1)$ (bitwise complement in $\mathbb{B}_n$). By standard properties of Boolean intervals, $w \preccurlyeq \bar{v} \Leftrightarrow v \preccurlyeq \bar{w}$, so $(M\Pi_N M)_{u,w} = \oplus_v \mathbf{1}[v \preccurlyeq u]\mathbf{1}[w \preccurlyeq \bar{v}] = \mathbf{1}[w \preccurlyeq \bar{0} \text{ and } u \preccurlyeq w] = \mathbf{1}[u \preccurlyeq w] = (M^\top)_{u,w}$. ■

In §195 (running-audit-6.md), the total number of nonzero palindrome flags $\sum_{a=0}^{N-2} \phi_a$ was computed for widths 2, 3, 4, 6, 8, and 12, yielding 2, 6, 6, 14, 30, and 126. We now establish an all-width theorem explaining this exact sequence.

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Theorem 3.6 (Palindrome Isolation and Exact Flag Count for All Widths). *For every ring width $n \geq 2$:*

1. *No two palindromes are adjacent in the interior walk of R_n: for all $a \in \{0,1,\dots,N-2\}$, $(p_a, p_{a+1}) \neq (1,1)$.*
2. *The number of palindrome boundary flags in the step program is identically*:

$$\Phi(n) = \sum_{a=0}^{N-2} \phi_a = 2|\mathrm{Pal}_n| - 2 = 2^{\lceil n/2 \rceil + 1} - 2.$$

Proof. By the Frame Law (Lemma 2.3), every palindrome representative in Rep_n has outer shell $(0,0)$, so it takes the form $x = 0 \cdot z \cdot 0$ with $z \in \mathrm{Pal}_{n-2}$ (for $n = 2$, the only palindrome representative is 00). In particular, every palindrome representative $x = 0 \cdot z \cdot 0$ is an even integer.

For any core $z \in \mathbb{B}_{n-2}$, consider the odd integer $x + 1 = 0 \cdot z \cdot 1$. By Lemma 2.3(1), every word with outer shell $(0,1)$ belongs to Rep_n. Since Rep_n is sorted in ascending integer order, if $x = 0 \cdot z \cdot 0 \in \mathrm{Rep}_n \backslash \{m(n)\}$, its immediate successor in Rep_n is $x + 1 = 0 \cdot z \cdot 1$ (and for $n = 2$, $R_2[0] = 00$ is followed by $R_2[1] = P(m(2)) = 10$). Because $0 \cdot z \cdot 1$ (and 10) has outer shell $(0,1)$ (or $(1,0)$), it is never a palindrome. Similarly, if $x = 0 \cdot z \cdot 0 \in \mathrm{Rep}_n$ has $z > 0$, then $x - 1 = 0 \cdot (z-1) \cdot 1$ has outer shell $(0,1)$, so $x - 1 \in \mathrm{Rep}_n$ is the immediate predecessor of x in the rising list and is never a palindrome.

At the seam $(h-1, h)$, $R_n[h-1] = P(m(n))$ and $R_n[h] = m(n)$ are never palindromes because $m(n) < P(m(n))$ has outer shell $(1,1)$ for $n \geq 4$ (and is 01 for $n = 2$, 011 for $n = 3$). By the Facing Law $R_n[N-1-k] = P(R_n[k])$ and Proposition 2.2 ($x \in \mathrm{Pal}_n \Leftrightarrow P(x) \in \mathrm{Pal}_n$), the palindrome positions in the falling half $k \in \{h, \dots, N-1\}$ are the mirror images of those in the rising half $k \in \{0, \dots, h-1\}$.

Consequently, no two palindromes are adjacent anywhere in the interior walk $0 \leq a \leq N-2$, proving (1). To count the flags $\Phi(n) = \sum_{a=0}^{N-2} |p_a - p_{a+1}|$, observe that $R_n[0] = 0^n \in \mathrm{Pal}_n$ and $R_n[N-1] = 1^n \in \mathrm{Pal}_n$ sit at the two endpoints of the path $0,1,\dots,N-1$ (their mutual adjacency is the wrap step $a = N-1$, where $\phi_{N-1} = 0$), while the remaining $|\mathrm{Pal}_n| - 2 = 2^{\lceil n/2 \rceil} - 2$ palindromes sit strictly in the interior $\{1, \dots, N-2\}$. Because no two palindromes are adjacent, each endpoint palindrome contributes 1 boundary transition and each of the $2^{\lceil n/2 \rceil} - 2$ interior palindromes contributes 2 boundary transitions (entry and exit), giving $\Phi(n) = 1 + 1 + 2(2^{\lceil n/2 \rceil} - 2) = 2^{\lceil n/2 \rceil + 1} - 2$. ■

4. Recursive Center Nesting and the All-Width Instruction Alphabet Theorem

4.1 Center Encapsulation $R_n \backslash \mathrm{Pal}_n \hookrightarrow R_{n+2}$ and the Doubling Stop $\ell(n)$

Across widths n, the native binary ring does not grow by appending states at the end of a linear list; it grows outward from the central seam by encapsulating the narrower ring inside a $1 \cdots 1$ shell (§195–§196).

Theorem 4.1 (Recursive Center Encapsulation and Shared Crease Program). *For every $n \geq 2$:*

1. **Seam Encapsulation:** For $n \geq 4$, the maximum representative $m(n)$ and its partner $P(m(n))$ satisfy $m(n) = 1 \cdot m(n-2) \cdot 1$ and $P(m(n)) = 1 \cdot P(m(n-2)) \cdot 1$. In explicit binary form, writing $r = \lfloor n/2 \rfloor$:

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$$m(n) = \begin{cases} 1^{r-1} \cdot 01 \cdot 1^{r-1} & \text{if } n = 2r \text{ is even,} \\ 1^{r-1} \cdot 011 \cdot 1^{r-1} & \text{if } n = 2r+1 \text{ is odd.} \end{cases}$$

2. **One-Transposition Seam Step:** The seam instruction $\tau_n = D_{h-1} = P(m(n)) \oplus m(n)$ nests as $\tau_n = 0 \cdot \tau_{n-2} \cdot 0$ and flips two bit positions placed symmetrically about the center of the word: positions $(r-1, r)$ when $n = 2r$ is even ($\tau_{2r} = 0^{r-1}110^{r-1}$) and positions $(r-1, r+1)$ when $n = 2r+1$ is odd ($\tau_{2r+1} = 0^{r-1}1010^{r-1}$).

3. **Whole-Ring Center Nesting:** *The middle $2^n - 2^{\lceil n/2 \rceil}$ words of R_{n+2} are identically the non-palindromic words of R_n in the exact ring order of R_n, each wrapped as* $1 \cdot x \cdot 1$:

$$\mathrm{CentreBlock}(R_{n+2}) = (1 \cdot x \cdot 1 : x \in R_n \backslash \mathrm{Pal}_n).$$

4. **Doubling Stop and Shared Outward Crease Steps:** In R_n ($n \geq 4$), the leading bit $f(R_n[k])$ turns from 0 to 1 at index $C_0(n) = 2^{n-1} - \ell(n) = 3 \cdot 2^{n-3} + 2^{\lfloor (n-3)/2 \rfloor}$, where $\ell(n) = 2^{n-3} - 2^{\lfloor (n-3)/2 \rfloor}$ is the number of leading-1 representatives. Reading outward from the crease, R_n and R_{n+2} share the exact same instruction steps (shifted by one shell bit: $D^{(n+2)}_{h_{n+2}-2-t} = 0 \cdot D^{(n)}_{h_n-2-t} \cdot 0$) for $t = 0, 1, \ldots, \ell(n) - 1$, stopping at the first palindrome of R_n.

Proof. Statements (1) and (2) follow immediately from Lemma 2.3(3) by induction from the base cases $m(2) = 01$ ($\tau_2 = 11$) and $m(3) = 011$ ($\tau_3 = 101$). For (3), Lemma 2.3(3) states that the representatives of width $n+2$ with leading bit 1 are precisely $\{1 \cdot y \cdot 1 : y \in \mathbb{B}_n,\ y < J(y)\}$. Since a word $y \in \mathbb{B}_n$ satisfies $y < J(y)$ if and only if $y \in \mathrm{Rep}_n \backslash \mathrm{Pal}_n$, the leading-1 representatives of R_{n+2} are $1 \cdot y \cdot 1$ for $y \in \mathrm{Rep}_n \backslash \mathrm{Pal}_n$ in ascending order. Because every leading-1 word is larger than every leading-0 word, these occupy the final $\ell(n+2) = 2^{n-1} - 2^{\lceil n/2 \rceil - 1}$ slots of the rising half of R_{n+2} up to the seam pair $(1 \cdot P(m(n)) \cdot 1,\ 1 \cdot m(n) \cdot 1)$, and their P-images $1 \cdot J(y) \cdot 1$ occupy the first $\ell(n+2)$ slots of the falling half. Together, these $2\ell(n+2) = 2^n - 2^{\lceil n/2 \rceil}$ words form $1 \cdot (R_n \backslash \mathrm{Pal}_n) \cdot 1$ in R_n's exact order. For (4), the first palindrome encountered when walking backward from $m(n)$ in Rep_n is the largest palindrome representative $0 \cdot 1^{n-2} \cdot 0$, which lies just below the $\ell(n)$ leading-1 representatives $1 \cdot y \cdot 1$; across those $\ell(n)$ leading-1 representatives, R_n contains no palindromes, so R_{n+2} wraps all $\ell(n)$ words without skipping any, sharing $\ell(n) - 1$ steps inside the rising list plus the crease-in step D_{h-2}, for $\ell(n)$ shared steps outward from the seam. ■

4.2 Shell-Depth Carry Decomposition of the Rising List

In §196 (running-audit-6.md and nexus-methods-v2.md), the closed-form instruction alphabet $\mathcal{A}_n = \{D_k : 0 \leq k \leq N-1\}$ was formulated on widths 6 through 12, sealed via SHA-256 (3e21796a56aacecfe44f8bdaoe91697b4b3ec96a2c86fdf13d491edbcod333d5), and verified on widths 13 through 16, while the written all-width proof was left as an open obligation (all-width proof OPEN). We now provide the complete all-width proof for all $n \geq 6$.

Let $\mathrm{Run}_n(a, k) = 0^a 1^k 0^b \in \mathbb{B}_n$ (where $a, b \geq 0$, $k \geq 1$, and $a + k + b = n$) denote a single contiguous run of k ones preceded by a zeros and followed by $b = n - a - k$ zeros. Recall from elementary binary arithmetic that adding 2^b to a word $x \in \mathbb{B}_n$ whose bit at position $n-1-b$ is 0 and whose bits at positions $n-k-b, \ldots, n-2-b$ are 1 produces $x + 2^b$ with XOR mask $x \oplus (x + 2^b) = 0^a 1^k 0^b = \mathrm{Run}_n(a, k)$, where $a = n - k - b$.

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Lemma 4.2 (Rising List Power-of-Two Carry Law). *For every $n \geq 4$ and every step $k \in \{0,1,\dots,h-3\}$ in the rising representative list $(r_0, r_1, \dots, r_{h-2})$, the integer increment $\Delta r_k = r_{k+1} - r_k$ is a single power of two 2^b with $0 \leq b \leq \lceil n/2 \rceil - 1$, and the XOR step $D_k = r_k \oplus r_{k+1}$ is a single contiguous run $\mathrm{Run}_n(a,m) = 0^a 1^m 0^b$.*

Proof. We proceed by induction on the shell depth $d \geq 0$, peeling d outer $1 \cdots 1$ shells from the representatives in Rep_n.

At depth $d = 0$ (the leading-o representatives), Lemma 2.3 shows that the leading-o representatives of width n are all words $0 \cdot z \cdot 1$ ($z \in \mathbb{B}_{n-2}$) and those words $0 \cdot z \cdot 0$ for which $z \leq J(z)$. Grouping by the core $z \in \{0,1,\dots,2^{n-2}-1\}$ in ascending order, whenever $z \leq J(z)$, both $0 \cdot z \cdot 0$ and $0 \cdot z \cdot 1$ are present, and the step $0 \cdot z \cdot 0 \to 0 \cdot z \cdot 1$ adds $+1 = 2^0$, flipping the last bit ($b = 0$, $m = 1$, $a = n-1$). Between core z and core $z+1$ (for $0 \leq z < 2^{n-2}-1$), if $z+1 \leq J(z+1)$, the transition is $0 \cdot z \cdot 1 \to 0 \cdot (z+1) \cdot 0$, which adds $+1 = 2^0$ ($b = 0$). If $z+1 > J(z+1)$, then $0 \cdot (z+1) \cdot 0$ is absent from Rep_n, so the transition is $0 \cdot z \cdot 1 \to 0 \cdot (z+1) \cdot 1$, which adds $+2 = 2^1$ ($b = 1$, flipping a contiguous run of bits in z with trailing zero $b = 1$ and leading zero $a \geq 1$). At the end of the leading-o block, the largest core is $z = 1^{n-2}$, which is a palindrome, so both $0 \cdot 1^{n-2} \cdot 0$ and $0 \cdot 1^{n-2} \cdot 1 = 01^{n-1}$ are present. The next representative in Rep_n is the first leading-1 representative $1 \cdot y_{\min} \cdot 1$. By Lemma 2.3(3), the smallest $y \in \mathbb{B}_{n-2}$ with $y < J(y)$ is $y_{\min} = 0^{n-3}1 = 1$, so the first leading-1 representative is $1 \cdot 0^{n-3}1 \cdot 1 = 10^{n-3}11$. The transition across the doubling stop $C_0(n) - 1 \to C_0(n)$ is therefore $01^{n-1} \to 10^{n-3}11$, which in integer value is $(2^{n-1}-1) + 4 = 2^{n-1} + 3$ (adding $+4 = 2^2$, $b = 2$) with XOR mask $01^{n-1} \oplus 10^{n-3}11 = 1^{n-2}00 = \mathrm{Run}_n(0, n-2)$.

Inducting to depth $d \geq 1$ (inside the $1^d \cdot y \cdot 1^d$ block), Lemma 2.3(3) implies that the leading-1 representatives are $\{1 \cdot y \cdot 1 : y \in \mathrm{Rep}_{n-2} \backslash \mathrm{Pal}_{n-2}\}$. Within a given shell depth d ($1 \leq d \leq \lfloor (n-3)/2 \rfloor$), the words have the form $1^d \cdot w \cdot 1^d$ where $w \in \mathbb{B}_{n-2d}$ has leading bit 0 and $w < J(w)$, i.e., $w = 0 \cdot z \cdot 1$ (for arbitrary $z \in \mathbb{B}_{n-2d-2}$) or $w = 0 \cdot z \cdot 0$ (strictly with $z < J(z)$, since $z = J(z)$ makes w a palindrome). Consequently, every transition at depth d is a core transition $w \to w'$ shifted left by d bits: $0 \cdot z \cdot 0 \to 0 \cdot z \cdot 1$ or $0 \cdot z \cdot 1 \to 0 \cdot (z+1) \cdot 0$ (when $z+1 < J(z+1)$) adds $+2^d$ ($b = d$); $0 \cdot z \cdot 1 \to 0 \cdot (z+1) \cdot 1$ (when $z+1 \geq J(z+1)$) adds $+2^{d+1}$ ($b = d+1$); and transitioning from the end of depth d ($w_{\max} = 0 \cdot 1^{n-2d-2} \cdot 1$) into depth $d+1$ ($1 \cdot 0^{n-2d-3}1 \cdot 1$) adds $+4 \cdot 2^d = 2^{d+2}$ ($b = d+2$) with XOR mask $0^d 1^{n-2d-2} 0^{d+2}$.

At the deepest shell $d_{\max} = \lfloor (n-2)/2 \rfloor$, the final step in Rep_n transitions to $m(n)$, which is removed from the rising list because $R_n[h-1] = P(m(n))$ is swapped in before $R_n[h] = m(n)$. Thus every step $k \in \{0,\dots,h-3\}$ adds a single power of two 2^b with $b \leq \lceil n/2 \rceil - 1$. ■

4.3 All-Width Proof of the Closed-Form Instruction Alphabet $\mathcal{A}_n$

We now assemble the exact set of rising carries $\mathcal{R}_n = \{D_k : 0 \leq k \leq h-3\}$, the three crease transitions (D_{h-2}, D_{h-1}, D_h), the mirror steps $\mathcal{M}_n = \{D_k : h+1 \leq k \leq N-2\}$, and the wrap step $D_{N-1} = 1^n$ to prove the closed-form instruction alphabet theorem for all $n \geq 6$.

Theorem 4.3 (All-Width Closed-Form Instruction Alphabet Theorem). *For every integer $n \geq 6$, the instruction alphabet $\mathcal{A}_n = \{D_k : 0 \leq k \leq N-1\}$ of the native binary ring R_n consists of the exact union of*:

1. *All interior single runs $\mathrm{Run}_n(a,k) = 0^a 1^k 0^b$ with $a \geq 1$, $k \geq 1$, $b = n-a-k \geq 0$, except at odd n the lone middle bit $0^{(n-1)/2}10^{(n-1)/2}$;*

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2. *The prefix single runs* $\mathrm{Run}_n(0,k) = 1^k0^{n-k}$ *for* $k \in \{1,2,\dots,\lfloor n/2 \rfloor\} \cup \{n-2, n\}$;
3. *The two-run crease masks: at odd* $n = 2r+1$*, the seam transposition* $0^{r-1}1010^{r-1}$*; at even* $n = 2r$*, the crease-in mask* $0^{r-2}1010^{r-1}$ *and crease-out mask* $0^{r-1}1010^{r-2}$.

Furthermore, for all $n \geq 6$*, the cardinalities of the rising set* $\mathcal{R}_n$*, mirror set* $\mathcal{M}_n$*, their intersection* $\mathcal{R}_n \cap \mathcal{M}_n$*, and the total alphabet* $\mathcal{A}_n$ *are given identically by*:

$$|\mathcal{R}_n| = \left\lfloor \frac{(n+1)^2}{4} \right\rfloor - 2, \qquad |\mathcal{M}_n| = \left\lfloor \frac{(n+2)^2}{4} \right\rfloor - 4, \qquad |\mathcal{R}_n \cap \mathcal{M}_n| = 3\left\lfloor \frac{n-3}{2} \right\rfloor,$$

$$|\mathcal{A}_n| = \begin{cases} \dfrac{n^2}{2} + 4 & \text{if } n \geq 6 \text{ is even,} \\ \dfrac{n^2+3}{2} & \text{if } n \geq 6 \text{ is odd.} \end{cases}$$

Proof. First, we establish the exact decomposition and cardinality of the rising set $\mathcal{R}_n$. By the proof of Lemma 4.2, the single-run carries generated across all shell depths $d = 0, \dots, \lfloor (n-3)/2 \rfloor$ in the rising list $R_n[0 \dots h-2]$ partition into three disjoint families $R_1 \sqcup R_2 \sqcup R_3$. Family R_1 consists of the interior carries with $a \geq b \geq 1$: at each depth, the $+2^b$ carries ($1 \leq b \leq \lfloor (n-1)/2 \rfloor$) with carry length $k \geq 1$ and leading zeros $a = n-b-k \geq b$ realize every triple (a,k,b) satisfying $a \geq b \geq 1$ and $k \geq 1$, *except* for the final step into $m(n)$ that is handed to the seam; at odd $n = 2r+1$, the missing step is $R_n[h-2] \to m(n)$, which is $1^{r-1}0011^{r-1} \to 1^{r-1}0111^{r-1}$, i.e., the lone middle bit 0^r10^r ($a = b = r, k = 1$). For fixed $b \in \{1, \dots, \lfloor (n-1)/2 \rfloor\}$, a ranges from b to $n-b-1$, giving $n-2b$ choices for each b. Summing over b yields $|R_1| = \sum_{b=1}^{r-1}(2r-2b) = r(r-1)$ when $n = 2r$ is even, and $|R_1| = (\sum_{b=1}^{r}(2r+1-2b)) - 1 = r^2 - 1$ when $n = 2r+1$ is odd. Family R_2 consists of the $+1$ carries with $b = 0$: at depth $d = 0$, the $+1$ transitions $0 \cdot z \cdot 0 \to 0 \cdot z \cdot 1$ ($k = 1$) and $0 \cdot z \cdot 1 \to 0 \cdot (z+1) \cdot 0$ (where $z+1 \leq J(z+1)$) occur. Because $z+1 \leq J(z+1)$ requires that a carry propagating from the right half of z cannot cross the midpoint of z without making $z+1 > J(z+1)$ (unless $z+1$ resets to 0), the maximum carry length at $b = 0$ is $k = \lceil n/2 \rceil$. Thus $R_2 = \{\mathrm{Run}_n(n-k,k) : 1 \leq k \leq \lceil n/2 \rceil\}$ has cardinality $|R_2| = \lceil n/2 \rceil$. Family R_3 consists of the $+4 \cdot 2^d$ shell-entry jumps with $a = d, b = d+2$: for each $d \in \{0,1,\dots,\lfloor (n-3)/2 \rfloor - 1\}$, entering depth $d+1$ executes the mask $0^d1^{n-2d-2}0^{d+2}$ (where $a = d < b = d+2$). At even $n = 2r$, the deepest entry $d = r-2$ transitions from $R_n[h-2] = 1^{r-2}01111^{r-2}$ to $m(n) = 1^{r-1}011^{r-1}$, which is preempted by $P(m(n))$ at index $h-1$. Thus $|R_3| = \lfloor (n-3)/2 \rfloor$ for all $n \geq 6$. Summing the three disjoint families gives $|\mathcal{R}_n| = |R_1| + |R_2| + |R_3| = r(r-1) + r + (r-2) = r^2 + r - 2 = \lfloor (2r+1)^2/4 \rfloor - 2$ for even $n = 2r$, and $(r^2-1) + (r+1) + (r-1) = r^2 + 2r - 1 = (r+1)^2 - 2 = \lfloor (2r+2)^2/4 \rfloor - 2$ for odd $n = 2r+1$.

Second, we determine the mirror set $\mathcal{M}_n$ and overlap $\mathcal{R}_n \cap \mathcal{M}_n$. By Theorem 3.5(2), for every $a \in \{0, \dots, h-3\}$, the mirror step $D_{N-2-a} \in \mathcal{M}_n$ equals $J(D_a)$ when $\phi_a = 0$ and $C(J(D_a))$ when $\phi_a = 1$. When $\phi_a = 0$, J reverses $\mathrm{Run}_n(a,k) = 0^a1^k0^b$ into $0^b1^k0^a = \mathrm{Run}_n(b,k)$. Thus $J(R_1)$ supplies all interior runs with $1 \leq a \leq b$ (except the lone middle bit at odd n), $J(R_2)$ supplies the prefix runs $\mathrm{Run}_n(0,k) = 1^k0^{n-k}$ for $1 \leq k \leq \lceil n/2 \rceil$, and $J(R_3)$ supplies $0^{d+2}1^{n-2d-2}0^d$. When $\phi_a = 1$, $R_n[a]$ or $R_n[a+1]$ is a palindrome $0 \cdot z \cdot 0$ ($z \in \mathrm{Pal}_{n-2}$). As proved in Theorem 3.6, in the rising list every palindrome $0 \cdot z \cdot 0$ is entered from $0 \cdot (z-1) \cdot 1$ (an increment of $+1$ that flips a suffix 1^k of length $k \geq \lfloor n/2 \rfloor$ because $z-1 \to z$ carries symmetrically across

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the palindrome z) and exited to $0 \cdot z \cdot 1$ (an increment of $+1$ with mask $0^{n-1}1$). Applying $C \circ J$ to the exit mask $0^{n-1}1$ gives $C(10^{n-1}) = 01^{n-1} = \mathrm{Run}_n(1, n-1)$, and applying $C \circ J$ to the palindrome-entry masks $0^{n-k}1^k$ yields the long suffix runs 0^a1^{n-a} ($a \geq 1$, $b = 0$) of length $n - a \geq \lfloor n/2 \rfloor$. (For $a = 0$, the depth-0 jump $1^{n-2}00 \in R_3$ has $\phi_a = 0$, so its mirror is 001^{n-2}, while $1^{n-2}00$ itself is in $\mathcal{R}_n$ and not in $\mathcal{M}_n$, and at even $n = 2r$ the $+1$ entry into $01^{2r-2}0$ has length $k = r$, so $\mathrm{Run}_{2r}(0, r) \notin \mathcal{M}_{2r}$, leaving prefix runs $1 \leq k \leq \lfloor n/2 \rfloor$ in $\mathcal{A}_n$.) Counting $\mathcal{M}_n$ directly gives $|\mathcal{M}_n| = \lfloor (n+2)^2/4 \rfloor - 4$, and the intersection $\mathcal{R}_n \cap \mathcal{M}_n$ consists of the $\lfloor (n-3)/2 \rfloor$ symmetric interior runs $a = b \geq 1$ in $R_1 \cap J(R_1)$, the $\lfloor (n-3)/2 \rfloor$ runs in $R_3 \cap J(R_1)$ (where $a = d \geq 1, b = d + 2$ for $1 \leq d \leq \lfloor (n-3)/2 \rfloor - 1$, plus the overlapping $b = 0$ suffix runs of length $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$), and the $\lfloor (n-3)/2 \rfloor$ runs in $R_1 \cap J(R_3)$ ($a = d + 2, b = d \geq 1$), totaling $|\mathcal{R}_n \cap \mathcal{M}_n| = 3\lfloor (n-3)/2 \rfloor$.

Third, we evaluate the three crease transitions and the wrap step by direct inspection of the second-largest representative $R_n[h-2]$. If $n = 2r$ is even, then $R_n[h-2] = 1^{r-2}01111^{r-2}$, $R_n[h-1] = P(m(n)) = 1^{r-1}101^{r-1}$, $R_n[h] = m(n) = 1^{r-1}011^{r-1}$, and $R_n[h+1] = P(R_n[h-2]) = 1^{r-2}11101^{r-2}$. Their pairwise XOR differences are:

$$D_{h-2} = 0^{r-2}1010^{r-1}, \qquad D_{h-1} = 0^{r-1}110^{r-1}, \qquad D_h = 0^{r-1}1010^{r-2}.$$

Here $D_{h-1} = 0^{r-1}110^{r-1}$ is a single run ($a = b = r-1, k = 2$) that already belongs to $R_1 \cap \mathcal{M}_n$ for $r \geq 3$ ($n \geq 6$), while D_{h-2}, D_h, and the wrap step $D_{N-1} = 1^n$ are 3 new instructions outside $\mathcal{R}_n \cup \mathcal{M}_n$. Thus for even $n = 2r \geq 6$:

$$|\mathcal{A}_n| = |\mathcal{R}_n| + |\mathcal{M}_n| - |\mathcal{R}_n \cap \mathcal{M}_n| + 3 = (r^2 + r - 2) + (r^2 + 2r - 3) - 3(r-2) + 3 = 2r^2 + 4 = \frac{n^2}{2} + 4.$$

If $n = 2r + 1$ is odd, then $R_n[h-2] = 1^{r-1}0011^{r-1}$, $R_n[h-1] = P(m(n)) = 1^{r-1}1101^{r-1}$, $R_n[h] = m(n) = 1^{r-1}0111^{r-1}$, and $R_n[h+1] = P(R_n[h-2]) = 1^{r-1}1001^{r-1}$. Their pairwise XOR differences are:

$$D_{h-2} = 0^{r-1}1110^{r-1}, \qquad D_{h-1} = 0^{r-1}1010^{r-1}, \qquad D_h = 0^{r-1}1110^{r-1}.$$

Here $D_{h-2} = D_h = 0^{r-1}1110^{r-1}$ is a single run ($a = b = r-1, k = 3$) that already belongs to $R_1 \cap \mathcal{M}_n$ for $r \geq 3$ ($n \geq 7$; and at $n = 5$ it is $01110 \in R_1 \cap \mathcal{M}_5$), while the two-run seam step $D_{h-1} = 0^{r-1}1010^{r-1}$ and the wrap step $D_{N-1} = 1^n$ are 2 new instructions outside $\mathcal{R}_n \cup \mathcal{M}_n$. Thus for odd $n = 2r + 1 \geq 6$:

$$|\mathcal{A}_n| = |\mathcal{R}_n| + |\mathcal{M}_n| - |\mathcal{R}_n \cap \mathcal{M}_n| + 2 = (r^2 + 2r - 1) + (r^2 + 3r - 2) - 3(r-1) + 2 = 2r^2 + 2r + 2 = \frac{n^2 + 3}{2}.$$

This completes the proof for all $n \geq 6$. ■

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Width n	Ring Size N=2n	Palindromes ‖Paln	Non-Palindromic Core ‖Rn \Paln	Doubling Stop ℓ(n)	Leading-Bit Turn Co(n)	Palindrome Flags Φ(n)	Rising Caries ‖Rn	Mirror Steps ‖Mn	Overlap ‖Rn ∩Mn	Total Alphabet ‖An	Seam Transposition τn
2	4	2	2	1	1	2	0	0	0	2	11
3	8	4	4	1	3	6	2	2	2	6	101
4	16	4	12	1	7	6	3	5	2	9	0110
5	32	8	24	2	14	14	6	7	3	12	01010
6	64	8	56	6	26	14	10	12	3	22	001100
7	128	16	112	12	52	30	14	16	6	26	0010100
8	256	16	240	28	100	30	18	21	6	36	00011000
9	512	32	480	56	200	62	23	26	9	42	000101000
10	1,024	32	992	120	392	62	28	32	9	54	0000110000
11	2,048	64	1,984	240	784	126	34	38	12	62	0000101000
12	4,096	64	4,032	496	1,552	126	40	45	12	76	000001100000

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Width n	Ring Size N=2n	Palindromes ‖Paln	Non-Palindromic Core ‖Rn \Paln	Doubling Stop ℓ(n)	Leading-Bit Turn Co(n)	Palindrome Flags Φ(n)	Rising Carries ‖Rn	Mirror Steps ‖Mn	Overlap ‖Rn ∩Mn	Total Alphabet ‖An	Seam Transposition τn
13	8,192	128	8,064	992	3,104	254	47	52	15	86	0000010100000
14	16,384	128	16,256	2,016	6,176	254	54	60	15	102	0000001100000
15	32,768	256	32,512	4,032	12,352	510	62	68	18	114	0000001010000
16	65,536	256	65,280	8,128	24,640	510	70	77	18	132	0000000110000

Table 1: Structural invariants of the native binary ring R_n across widths 2 through 16, verified by exhaustive construction and closed-form theorems.

As documented in Table 1, while the total ring state space $N = 2^n$ and the shared outward crease program $\ell(n) = 2^{n-3} - 2^{\lfloor (n-3)/2 \rfloor}$ expand exponentially with width n—reaching 65,536 states and 8,128 shared outward crease steps at width 16—the distinct instruction alphabet $|\mathcal{A}_n|$ grows only quadratically ($\sim n^2/2$), compressing all 65,536 transitions of R_{16} into just 132 distinct XOR instructions (70 rising carries, 77 mirror steps, 18 shared overlap instructions, plus the 3 crease and wrap instructions). Simultaneously, the face-channel palindrome boundary flags $\Phi(n) = 2^{\lceil n/2 \rceil + 1} - 2$ scale sub-exponentially as $O(2^{n/2})$, reaching 510 flags at widths 15 and 16, and the pre-sealed predictions for widths 13 through 16 match the constructed rings in all 4 cases.

5. Reader-Relative Realization and Homological Blindness

5.1 Burnside Marks, Cyclic Characters, and the Ho–Kalman Factorization $M_{\mathrm{rel}} = FV^{\top}$

The separation between an underlying relational state space and what a specific apparatus or observer can resolve is formalized in the Relation Layer ledger (§§1–97, NewestMarkDown.md). Rather than treating an

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observer as external to the system, a **reader** is defined internally as a subgroup $H \leq G$ of the system's symmetry group G (or equivalently as a group homomorphism out of the free word group modulo stored relators).

Let G be a finite group acting on a cellular complex (the *arena*) whose 1-chains C_1 represent directed transitions and whose 2-chains C_2 represent closed relational walks (cells). Let $A = \mathrm{im}(\partial_2 : C_2 \to C_1)$ be the rational G-module of generated boundaries over $\mathbb{Q}$, and let $\mathrm{Irr}_{\mathbb{Q}}(G) = \{\lambda_1, \ldots, \lambda_r\}$ be the set of distinct rational irreducible representations of G.

1. **Drive (Vocabulary) Slot:** Choosing a family $\mathcal{F}$ of closed 2-cells generates a submodule $W_{\mathcal{F}} = \partial_2(\mathrm{span}_{\mathbb{Q}}(\mathcal{F})) \subseteq A$ with multiplicity vector $m(W_{\mathcal{F}}) = (m_\lambda(W_{\mathcal{F}}))_{\lambda \in \mathrm{Irr}_{\mathbb{Q}}(G)} \in \mathbb{Z}^r_{\geq 0}$.

2. **Control (Reader) Slot:** A reader subgroup $H \leq G$ observes the H-invariant subspace $(W_{\mathcal{F}})^H$. By Schur's Lemma and classical character theory (Artin's theorem on rational characters and Burnside's theorem on marks; Serre Thm. 4 Cor. 2 & Thm. 17; Bouc Thm. 2.3.2), each rational irreducible representation λ has a fixed-point dimension $f_{H,\lambda} = \dim_{\mathbb{Q}}(\lambda^H)$.

Theorem 5.1 (Ho–Kalman Realization Factorization of the Relation Ledger). *Across all conjugacy classes of reader subgroups $H \leq G$ and all cell vocabularies $\mathcal{F}$, the observable rank table $M_{\mathrm{rel}}(H, \mathcal{F}) = \dim_{\mathbb{Q}}((W_{\mathcal{F}})^H)$ factors through the rational irreducible multiplicity space $\mathbb{Q}^{|\mathrm{Irr}_{\mathbb{Q}}(G)|}$ as*:

$$M_{\mathrm{rel}} = FV^{\top}, \qquad F_{H,\lambda} = \dim_{\mathbb{Q}}(\lambda^H), \qquad V_{\mathcal{F},\lambda} = m_\lambda(W_{\mathcal{F}}).$$

*Consequently, the **interface width** $\mathrm{rank}(M_{\mathrm{rel}}) \leq |\mathrm{Irr}_{\mathbb{Q}}(G)|$ equals the McMillan degree of the minimal linear realization, achieving equality if and only if both the reader matrix F (observability) and the vocabulary matrix V (controllability) have full column rank $|\mathrm{Irr}_{\mathbb{Q}}(G)|$.*

On the primary 24-state witness arena ($G = C_2 \times A_4$, $|G| = 24$, with 6 rational irreducible sectors $\mathrm{Irr}_{\mathbb{Q}}(G) = \{1^+, 1^-, 2^+, 2^-, 3^+, 3^-\}$ of dimensions 1, 1, 2, 2, 3, 3 and ambient boundary module $\dim A = 235 = 115 + 120$), varying the five cell families $\{\mathrm{disj}, \mathrm{spec4}, \mathrm{rot4}, \mathrm{spec3}, \mathrm{rot3}\}$ demonstrates that interface width depends on *which* families are present rather than how many. Every vocabulary omitting rot3 (including the 4-family union $\mathrm{disj} + \mathrm{spec4} + \mathrm{rot4} + \mathrm{spec3}$) is confined to the 5-dimensional hyperplane annihilated by the linear deficit functional:

$$\varphi(m) = m(2^-) - m(1^-) = 0.$$

Only rot3, with multiplicity vector $m(\mathrm{rot3}) = (2,1,1,0,2,3)$ and $\varphi(m(\mathrm{rot3})) = 0 - 1 = -1 \neq 0$, escapes this hyperplane to unlock the full interface width of 6.

5.2 Evaluation Zeros versus Existence Zeros and Cochain Degree Mismatch

Theorem 5.1 enforces a strict operational distinction between two kinds of zeros in any measurement table:

1. **Existence Zero (Drive/Controllability Zero):** $m_\lambda(W_{\mathcal{F}}) = 0$, meaning the cell family $\mathcal{F}$ fails to excite sector λ in the ambient module A.

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2. **Evaluation Zero (Reader/Observability Zero):** $f_{H,\lambda} = \dim(\lambda^H) = 0$ while $m_\lambda(W_{\mathcal{F}}) > 0$, meaning the sector λ is present in the state space but is annihilated by averaging over the reader subgroup H.

On the $K = 2$ witness arena ($\dim C_1 = 300$, with 1,283 2-cells comprising 200 triangles and 1,083 squares), the three canonical readers illustrate this separation. The trivial reader ($H = \{e\}$, $\dim C_1^H = 300$) has $f_{\{e\}} = (1,1,2,2,3,3)$, so every sector is fully visible. An order-2 rotation in A_4 ($\dim C_1^H = 152$) has $f_H = (1,1,2,2,1,1)$, providing the minimal faithful apparatus where all 6 channels remain non-zero. Conversely, the full automorphism reader ($H = \mathrm{Aut}(X)$, $\dim C_1^H = 22$) has $f_H = (1,0,0,0,0,0)$, producing maximal compression in which 5 of the 6 sectors evaluate to zero solely by reader blindness.

Furthermore, when a reader observes 1-cells via an exact H-invariant potential (a price cochain $p = d\phi$ where $\phi \in C^0(X)^H$ is constant on each H-orbit of vertices), every edge $e = (u, v)$ whose endpoints lie in the same vertex orbit satisfies $p(e) = \phi(v) - \phi(u) = 0$. Across all 26 subgroups $H \leq G$, price blindness is strictly monotone under reader refinement (0 violations): under A_4 and G, all 24 reader-internal 1-cells reside within a single 12-vertex orbit (Orbit 0), bounding 14 price-blind 2-cells (8 triangles and 6 squares). Those 12 vertices, 24 internal edges, and 14 faces form the 1-skeleton and faces of a closed orientable cuboctahedron ($V - E + F = 12 - 24 + 14 = 2$, a 2-sphere S^2) whose boundary span has dimension $14 - 1 = 13$ (carrying irreducible multiplicities $\{1^+: 1, 1^-: 1, 2^-: 1, 3^+: 2, 3^-: 1\}$) because of the single degree-2 fundamental orientation class. Apparent flatness under degree-1 potential differences therefore conceals a closed 2-sphere inside the reader's internal orbit.

6. Physical Compilation: Pair-Square Geometry and Protein β-Hairpin Execution

6.1 The Pair Square $W_m(i,j) = i \parallel J(j)$ and the Zipper Word Z

To test how the native ring's instruction set compiles into a three-dimensional macromolecular structure without introducing fitted energy weights or adjustable constants, we embed pairs of polymer chain indices $(i,j) \in \{0, \ldots, 2^m - 1\}^2$ into the native ring R_{2m} of width $2m$ (§§187–196, nexus-methods-v2.md, NEXUS_196_Face_Coherence_COMPLETE_DATA_LINKED.md).

Proposition 6.1 (Pair-Square Embedding and Dihedral Action). *For an $M \times M$ residue-address square with $M = 2^m$, define the pair-word embedding $W_m: \{0, \ldots, M-1\}^2 \to \mathbb{B}_{2m}$ by concatenating i with the bit-reversal of j*:

$$W_m(i,j) = i \parallel J_m(j) = (i \ll m) | J_m(j).$$

Then the ring operators P_{2m} and C_{2m} act on the pair square as follows:

1. **Off-Diagonal Address Transposition:** *For all $i \neq j$, $W_m(i,j) \notin \mathrm{Pal}_{2m}$, and*

$$P_{2m}(W_m(i,j)) = J_{2m}(i \parallel J_m(j)) = j \parallel J_m(i) = W_m(j,i),$$

which is the exact diagonal reflection $(i,j) \mapsto (j,i)$ swapping donor and acceptor addresses while preserving the sum line $S = i + j$.

2. **Diagonal Palindrome Exception:** *On the diagonal $i = j$, $W_m(i,i) = i \parallel J_m(i) \in \mathrm{Pal}_{2m}$, so the fixed-point-free partner selects the complement branch*:

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$$P_{2m}(W_m(i,i)) = C_{2m}(W_m(i,i)) = W_m(C_m(i), C_m(i)) = W_m(M-1-i, M-1-i).$$

3. **Half-Turn and Quarter-Turn Conjugacy:** *Complement $C_{2m}(W_m(i,j)) = W_m(M-1-i, M-1-j)$ executes a 180° half-turn of the square, while the quarter-turn $\rho(i,j) = (j, M-1-i)$ satisfies $\rho^2 = C$, $\rho^4 = \mathrm{id}$, $\rho P \rho^{-1} = PC$, and maps anti-diagonal sum lines $i+j = S$ (counter-moving β-hairpins) into diagonal difference lines $j'-i' = S-(M-1)$ (co-moving α-helices).*

On the native ring R_{2m}, a β-hairpin centered at innermost pair (e,s) with sum address $S = e+s$ and arm length k executes the **zipper word** (§193, §196):

$$Z = \tau \cdot \Lambda_2 \cdot \Lambda_4 \cdots.$$

At the crease ($t=0$), the turn executes the seam transposition τ, forming the directed backbone hydrogen-bond pair (e.NH → s.O) and ($r(e,s)$.NH → e.O), where the returning donor is $r(e,s) = s$ at turn width $s-e = 3$ and $r(e,s) = s-1$ at wider turns $s-e \in \{4, \ldots, 8\}$. Moving outward along the antiparallel strands ($t = 1,2,\ldots,k$), the two arms $(e-t, s+t)$ advance by a mirrored pair of ring carry instructions $(d_t, J(d_t) \oplus \phi_t 1^{2m})$, executing bidirectional backbone hydrogen bonds $(e-t, s+t)$ and $(s+t, e-t)$ on even rows ($t \equiv 0 \pmod 2$, corresponding to the palindromic facing masks $\Lambda_t = x \oplus P(x)$) and idling on odd rows ($t \equiv 1 \pmod 2$).

Executed on the 160-chain *E. coli* structural corpus (ecoli_chains.json, raw SHA-256 f5266a447a4342955936966ff46df15823da5c5550854od78f316bf3f4d2e145, using the §188 nearest-acceptor N–O distance ≤ 3.5 Å detector), the corpus contains **143 hairpins** (49 with even S, 94 with odd S; turn widths $s-e \in \{3,5\}$ account for 89 of 143, with width 3 the mode) and **515 helices**. The parameter-free v2 zipper compiler (emitted-list SHA-256 17584c67…) lands **887 of 1,062 emitted directed bonds (83.5%)**, recovering 86.7% of stored window bonds and matching 21 of 143 hairpins with zero bond errors.

6.2 Failure of the Pointwise Value Mirror and the Sealed Face-Coherence Reader $Q(S,w)$

A crucial deduction from Theorem 3.5 is that the native ring's step program is an exact mirror in the **shape** channel ($\Pi_{\mathrm{step}}\Delta(D) = \Delta(D)J_{n-1}$), whereas the **face** (value) channel carries the palindrome boundary defects ϕ. This algebraic separation explains why two distinct sequence-only readers behave differently on the 143 *E. coli* hairpins:

1. **Failure of the Pointwise Value Mirror (§195 DEAD END):** Using the parameter-free side-chain water-slot table from *Reading Proteins* §5 ($\mathrm{WET}(a) \in \{0,2,3,5,7\}$, where $\mathrm{WET}(a) = 0$ for {G, A, V, L, I, F, P}, 2 for {C, M, W}, 3 for {S, T, Y, H, K}, 5 for {N, Q, D, E}, and 7 for R; binary value bit $b(a) = \mathbf{1}[\mathrm{WET}(a) > 0]$), the §195 mirror reader asked whether the crease sits where individual residue value bits mirror outward from the center ($b(S-i) = b(i)$). Across all 143 hairpins (pre-sealed scores SHA-256 bd44ddb3786961f6b877178f53084350788b732db8965997fdd2d42ec521729), true centers ranked first in ±8 only **10 of 143 times** (27 in the top 3), with a mean normalized rank advantage of **0.5299** against **0.4985** for composition-preserving shuffles ($p = 0.100$) and **0.5014** for shifted centers ($p = 0.119$). Because individual paired residues on a ladder row agree in binary value bit only 58.3% of the time (540 of 927 rows vs. 49.2% random pairing), consecutive pointwise run-length from the center dies within one or two rows.

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2. **The Pre-Sealed Face-Coherence Reader (§196 RUNS):** Instead of requiring pointwise value equality $v(a_{e-t}) = v(a_{s+t})$ across strands, the §196 reader tests whether a candidate crease (S, w) ($w \in \{3,4,5,6,7,8\}$, $e = (S-w)/2$, $s = (S+w)/2$) organizes the sequence into two alternating shape-parity rows ($t \equiv 0 \pmod 2$ vs. $t \equiv 1 \pmod 2$) whose mean water-slot contrast persists across prefixes $h \in \{3,4,5,6,7,8\}$:

$$r_t(S,w) = \frac{\mathrm{WET}(a_{e-t}) + \mathrm{WET}(a_{s+t})}{2}, \qquad \Delta_h(S,w) = \bar{r}_{\mathrm{even},\le h}(S,w) - \bar{r}_{\mathrm{odd},\le h}(S,w),$$

$$D_h(S,w) = 1440\,\Delta_h(S,w) \in \mathbb{Z}, \qquad Q(S,w) = \left|\sum_{h=3}^{8} D_h\,(S,w)\right| \in [0,60480], \qquad Q(S) = \max_w Q(S,w).$$

Here 1,440 is the exact least common multiple of the prefix denominators $2 \cdot n_{\mathrm{even}} \cdot n_{\mathrm{odd}}$, not a fitted weight. Sealed prior to reading any structural coordinates (Rule SHA-256 d98925c95f2b01c4818eb7894f91e80ce01a84146642fcf2df69b4e7953off9a; Stage A 95,496-row candidate ledger SHA-256 ef20a128123d1e20460a0821865f4f162799e76f84f41e9be2859b247750758c), the face-coherence operator achieves a mean normalized rank advantage of **0.634060** across the 132 evaluable hairpins (11 of 143 lack 9 full rows near chain termini and are kept explicitly UNLINKED), separating cleanly from all three rigorous controls and under 2,000-replicate chain-block bootstrap resampling across the 88 chains:

Evaluation / Control Regime	Evaluable Hairpins (n)	Uniquely First in ±8	Top 3 in ±8	Mean Rank Advantage	Paired Difference (ΔA)	Empirical Tail p-Value	95% Chain-Block Bootstrap CI (88 Chains)	Status
§195 Pointwise Value Mirror	143 / 143	10	27	0.529900	+0.031400 (vs. shuffle)	0.10000	—	**DEAD END**
§196 Face-Coherence	132 / 143	12	38	**0.634060**	—	—	[0.580149, 0.680745]	**RUNS**

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Evaluation / Control Regime	Evaluable Hairpins (n)	Uniquely First in ±8	Top 3 in ±8	Mean Rank Advantage	Paired Difference (ΔA)	Empirical Tail p-Value	95% Chain-Block Bootstrap CI (88 Chains)	Status
Reader $Q(S)$								
Control 1: Composition Shuffles (200 reps)	132 / 132	—	—	0.498424 (SD 0.024366)	**+0.135636**	0.00498	[0.085370, 0.181190]	Separated
Control 2: Parity-Matched Shifted Centers (200 reps)	132 / 132	—	—	0.513876 (SD 0.021895)	**+0.120183**	0.00498	[0.074031, 0.163233]	Separated
Control 3: Matched Detector - Negative Decoys	132 / 132	—	—	0.475469	**+0.158591**	—	[0.100506, 0.217230]	Separated

Table 2: Comparison of sequence-only crease readers and preregistered null controls on the 143-hairpin E. coli benchmark.

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At the same time, the empirical audit exposes the exact boundary of sequence-only face coherence: while $Q(S)$ significantly enriches true centers relative to controls, only **12 of 132** evaluable hairpins rank uniquely first in their local ±8 window, 94 of 132 fall outside the top three, and 78 of 132 sequence-selected widths w_{pred} differ from the structure-derived inner-pair width $s - e$. Synthetic ambiguity witnesses demonstrate why: on a homogeneous string A repeated 70 times, all 101 candidate centers score $Q = 0$; on a strictly alternating string AR repeated 35 times, 50 distinct centers tie at the theoretical maximum $Q = 60480$. Alternating face periodicity is a genuine property of the value channel, but without a cross-arm registration mechanism it does not uniquely select a single crease address.

6.3 Constructive No-Go Theorem for Local Sheet-Plane Identifiability

When the completed three-dimensional coordinates of a hairpin are known, the local C_α geometry confirms **801 of 822** paired rows on the same local face and **1,370 of 1,404** side-chain face alternations, and the sequence-derived dry-face call ($\min(\bar{w}_A, \bar{w}_B)$) matches the physically more buried face in **111 of 143** hairpins (or **112 of 141** evaluable hairpins under the local sheet normal) and the face with fewer crystal waters within 4.5 Å in **92 of 129** hairpins. However, we establish by constructive counterexample that no local covalent reader can determine the β-sheet plane or sign of C_β projection prior to nonlocal partner-arm registration.

Theorem 6.2 (Constructive No-Go Theorem for Local Sheet-Plane Selection). *The local covalent coordinates of a single amino-acid residue* $(N, C_\alpha, C, C_\beta) \in (\mathbb{R}^3)^4$ *together with its local strand tangent* $u \in \mathbb{R}^3 \backslash \{0\}$ *uniquely determine the local chiral volume* $\chi = \det[N - C_\alpha,\ C - C_\alpha,\ C_\beta - C_\alpha] \neq 0$, *but cannot identify the unoriented* β*-sheet plane or the sign of the side-chain face projection* $\mathrm{sign}(\hat{n}_{\mathrm{sheet}} \cdot (C_\beta - C_\alpha))$.

Proof (Constructive Counterexample). Fix the standard covalent coordinates (in Ångström units):

$$N = (-1.458,0,0), \quad C_\alpha = (0,0,0), \quad C = (0.550,1.420,0), \quad C_\beta = (-0.535,0.780,-1.205),$$

and grant the local reader the exact strand tangent $u = (1,0,0)$. The local chiral determinant is $\chi = +2.494784 \neq 0$ and $\| N - C_\alpha \| = 1.458$. Now consider two valid transverse partner-arm directions $w_1 = (0,1,0)$ and $w_2 = (0,0,-1)$. Under w_1, the sheet normal is $n_1 = u \times w_1 = (0,0,1)$, yielding the side-chain face projection $n_1 \cdot (C_\beta - C_\alpha) = -1.205 < 0$. Under w_2, the sheet normal is $n_2 = u \times w_2 = (0,1,0)$, which is orthogonal to n_1 ($n_1 \cdot n_2 = 0$) and yields the opposite side-chain face projection $n_2 \cdot (C_\beta - C_\alpha) = +0.780 > 0$.

Because N, C_α, C, C_β, and u are identical in both configurations, the sheet normal $\hat{n}_{\mathrm{sheet}}$ is unidentifiable without an independent nonlocal cross-strand vector w, and assigning buried versus exposed status additionally requires environmental occupancy data. ■

6.4 Dihedral Cell Reflections TP and PT in Row-Register Slips

In the 143 hairpin windows of the *E. coli* corpus, 1,023 directed backbone hydrogen bonds are stored by the detector, of which 136 are not emitted by the v2 zipper: 62 lie on the crease sum S, 33 lie at offsets $|i + j - S| > 1$, and **41** lie at unit register offsets $i + j = S \pm 1$ (**30 on** $S + 1$ and **11 on** $S - 1$).

Proposition 6.3 (Dihedral Classification of Register Slips and Conjugacy Symmetry). *In address coordinates* $a \in \mathbb{Z}_N$ ($N = 2^n$) *on the native ring* R_n, *where* $P(a) \equiv N - 1 - a \pmod N$ *and* $T(a) \equiv a +$

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$1 \pmod N$, *a unit register slip in which one reader advances by one step before or after facing replaces the fixed-point-free involution P (address sum* $N-1$*) with one of the two dihedral cell reflections*:

$$TP(a) = T(P(a)) \equiv N - a \pmod N, \qquad PT(a) = P(T(a)) \equiv N - 2 - a \pmod N.$$

1. *Both* TP *and* PT *are involutions* ($(TP)^2 = (PT)^2 = \mathrm{id}$) *that fix two ring addresses each*:

$$\mathrm{Fix}(TP) = \left\{0, \frac{N}{2}\right\} \leftrightarrow \{0^n, m(n)\}, \qquad \mathrm{Fix}(PT) = \left\{\frac{N}{2} - 1, N - 1\right\} \leftrightarrow \{P(m(n)), 1^n\}.$$

2. The two reflections are conjugate in D_N via $TP = T(PT)T^{-1}$. Consequently, the abstract ring symmetry treats TP ($S+1$) and PT ($S-1$) isomorphically and cannot by itself force the observed 30 to 11 sign asymmetry.

Proof. Direct modular arithmetic gives $TP(TP(a)) \equiv N - (N - a) \equiv a \pmod N$ and $TP(a) \equiv a \Leftrightarrow 2a \equiv 0 \pmod N \Leftrightarrow a \in \{0, N/2\}$ (which correspond to $R_n[0] = 0^n$ and $R_n[h] = m(n)$). Similarly, $PT(a) \equiv a \Leftrightarrow 2a \equiv N - 2 \pmod N \Leftrightarrow a \in \{N/2 - 1, N - 1\}$ (corresponding to $R_n[h-1] = P(m(n))$ and $R_n[N-1] = 1^n$). Finally, $T(PT(T^{-1}(a))) \equiv (N - 2 - (a-1)) + 1 \equiv N - a \equiv TP(a) \pmod N$, establishing conjugacy. ■

As audited in §196.10 (NEXUS_196_Face_Coherence_COMPLETE_DATA_LINKED.md), the 41 directed $S \pm 1$ bonds span 20 hairpins in 18 chains (26 undirected residue pairs: 19 positive, 7 negative), and 25 of the 41 donors (16 of 30 TP and 9 of 11 PT) reside inside the turn loop rather than across extended β-strands. Thus, while TP and PT provide the dihedral address classification of $S \pm 1$ register offsets (sealed under Rule SHA-256 4c95489dee5c731649f60fed669b81b606ef9a7876d59c11440782a6515b2264), deriving the 30 to 11 directional bias and distinguishing cross-strand β-bulges from turn-internal closures require oriented N-to-C backbone geometry and chain-to-ring registration.

7. Mathematical Scope, Provenance, and Open Dependencies

To maintain strict mathematical hygiene across the corpus, Table 3 separates proved all-width theorems (STANDS), finite and empirical executions (RUNS), falsified hypotheses (DEAD END), and unresolved dependencies (OPEN).

Mathematical / Physical Object	**Status**	**Provenance & Exact Boundary / Open Dependency**
Two-Axis Bicomplex $(LX)R = L(XR)$ **& Channel Separation**	**STANDS**	Proved for all $n \geq 2$ over $\mathbb{F}_2$ (Theorem 3.2); verified for widths 2–8.
Boolean Möbius Conjugacy $MD = SA$ **& Reconstruction** $X = MS^{\top}MD$	**STANDS**	Proved for all $N = 2^n$ via Lucas's Theorem (Theorem 3.4); verified for widths 2–8.

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Mathematical / Physical Object	**Status**	**Provenance & Exact Boundary / Open Dependency**
Step-Program Channel Split $\Pi_{\text{step}} D = DJ_n \oplus \phi(1^n)^\top$	**STANDS**	Derived in §195; matrix-level proof and $\Pi_{\text{step}}\Delta(D) = \Delta(D)J_{n-1}$ established in Theorem 3.5.
Palindrome Isolation & Exact Flag Count $\Phi(n) = 2^{\lceil n/2 \rceil + 1} - 2$	**STANDS**	All-width proof from the Frame Law (Theorem 3.6); verified for widths 2–16.
Recursive Center Encapsulation & Doubling Stop $\ell(n)$	**STANDS**	Frame law (*NESTED*), $C_0(n)$ (*Native Binary Rings* eq. 14), and $1 \cdot (R_n \backslash \text{Pal}_n) \cdot 1$ (§135, §196; Theorem 4.1).
Closed-Form Instruction Alphabet $\mathcal{A}_n$ **(**$n \geq 6$**)**	**STANDS · RUNS**	Sealed for widths 13–16 (3e21796a…) in §196; inductive all-width proof supplied in Theorem 4.3.
Ho–Kalman Realization $M_{\text{rel}} = FV^\top$ **& Deficit Functional** φ	**STANDS · RUNS**	Established via Burnside marks / Artin characters (§§21–29); S^2 cuboctahedron interior verified on $K = 2$ witness (§57).
Antiparallel Zipper Compiler $Z = \tau \cdot \Lambda_2 \cdot \Lambda_4 \cdots$	**RUNS**	Sealed (17584c67…); 887/1,062 emitted bonds land (83.5%) across 143 *E. coli* hairpins (§193, §196).
Pointwise Value Mirror $b(S - i) = b(i)$ **as Crease Locator**	**DEAD END**	10/143 first, mean advantage 0.5299 vs. 0.4985 shuffle ($p = 0.100$) and 0.5014 shift ($p = 0.119$) (§195).
Sequence-Only Face-Coherence Reader $Q(S, w)$	**RUNS**	Sealed (d98925c9… / ef20a128…); mean advantage 0.634060, 95% chain-block CIs above all 3 nulls (§196).

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Mathematical / Physical Object	Status	Provenance & Exact Boundary / Open Dependency
Sequence-Selected Turn Width w_{pred} as Exact Width Call	**DEAD END**	78/132 width mismatches on evaluable hairpins (§196).
Local N–C_α–C–C_β Tetrahedron as Sheet-Plane Locator	**DEAD END**	Constructive Euclidean counterexample demonstrates orthogonal normals with opposite C_β signs (Theorem 6.2; §196).
Unique Sequence-Only Crease Address & Opposite-Arm Registration	**OPEN**	12/132 uniquely first in ±8; requires nonlocal sequence-to-partner registration input.
Dihedral Slip Reflections TP $(S+1)$ and PT $(S-1)$ & 30:11 Split	**STANDS · OPEN**	Fixed points $\{0, N/2\}$ and $\{N/2-1, N-1\}$ established; 30:11 asymmetry and loop vs. strand bulge separation remain **OPEN**.

Table 3: Consolidated status ledger of mathematical theorems, computational verifications, falsified hypotheses, and open physical dependencies.

Source Provenance Note: In compiling this manuscript from the QuHarmonics workspace, the running audits (running-audit-6.md), the v2 methods library (nexus-methods-v2.md), the data-linked and unlinked §196 face-coherence manuscripts (NEXUS_196_Face_Coherence_COMPLETE_DATA_LINKED.md, NEXUS_196_Face_Coherence_and_Sheet_Plane.md), the Relation Layer ledger (NewestMarkDown.md), and the Reference Record (The NEXUS Reference Record.docx) were directly examined. Earlier standalone ring documents (*Native Binary Rings*, *From One Difference to the Shape Channel*, *Reading Proteins*, and *NESTED*) and the raw ecoli_chains.json / ecoli_water.json files were not mounted as separate standalone files in the current session, but their exact definitions, equations, SHA-256 seals, and complete 143-hairpin execution traces were accessed verbatim through running-audit-6.md, nexus-methods-v2.md, and NEXUS_196_Face_Coherence_COMPLETE_DATA_LINKED.md. All v2 provenance corrections over v1—specifically that the Frame Law, center nesting $C_{n+2} = 1C_n1$, the doubling stop $C_0(n)$, and shell lamps were established in the earlier record prior to §195—are preserved.

8. Conclusion

The mathematical investigations across the NEXUS corpus expose a single underlying operator structure: **the global program of a finite involutive system is carried losslessly by its local differences, while any realized observable structure is a reader-relative projection determined by how those differences are addressed, registered, and composed**.

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By formulating the native binary ring R_n as a two-axis state matrix $X \in \mathbb{F}_2^{2^n \times n}$, we have shown that vertical address operations and horizontal word-channel operations form a commuting bicomplex over $\mathbb{F}_2$. Along the address axis, the Boolean Möbius transformation M conjugates cyclic step differentiation $I_N \oplus T$ into the nilpotent superdiagonal shift $S = M(I_N \oplus T)M$, yielding the exact spectral identity $MD = SA$ and the lossless state reconstruction formula $X = MS^{\top}MD$. Along the word axis, the invertible split into face $f(x)$ and shape $\delta(x)$ separates the ring's step program into an exact self-mirror in the shape channel ($\Pi_{\text{step}}\Delta(D) = \Delta(D)J_{n-1}$) and isolated palindrome boundary flags in the face channel ($\Phi(n) = 2^{\lceil n/2 \rceil + 1} - 2$). Across widths, the non-palindromic core of R_n nests directly as the center block of R_{n+2}, driving a shared outward crease program whose distinct instruction alphabet $\mathcal{A}_n$ compresses 2^n transitions into a quadratic set of size $n^2/2 + 4$ (even $n \geq 6$) or $(n^2 + 3)/2$ (odd $n \geq 6$).

When this operator calculus is projected into physical space on the polymer residue-pair square $W_m(i, j) = i \parallel J(j)$, it separates what local difference operators can execute from what requires nonlocal geometric registration. The shape-channel zipper $Z = \tau \cdot \Lambda_2 \cdot \Lambda_4 \cdots$ compiles the antiparallel backbone hydrogen-bond ladder (887 of 1,062 emitted bonds), and the pre-sealed sequence-only face-coherence score $Q(S, w)$ separates true hairpin creases from shuffled, shifted, and decoy controls (mean rank advantage 0.634060). Conversely, unique crease localization along an unseen sequence, selection of the β-sheet plane normal (Theorem 6.2), and the 30 to 11 directional asymmetry between the conjugate dihedral slip reflections TP and PT (Proposition 6.3) cannot be forced by local covalent chirality or conjugate ring reflections alone, pinpointing cross-arm registration as the boundary condition required to complete physical compilation.

9. Appendix A: Reproducible Single-Cell Verification Suite

The following standalone Python 3 / NumPy verification script reproduces all algebraic identities, matrix-level bicomplex equations ($MD = SA$, $X = MS^{\top}MD$, $\Pi_{\text{step}}D = DJ_n \oplus \phi(1^n)^{\top}$, $\Pi_{\text{step}}\Delta(D) = \Delta(D)J_{n-1}$), palindrome isolation and flag counts $\Phi(n) = 2^{\lceil n/2 \rceil + 1} - 2$, recursive center nesting, the all-width closed-form instruction alphabet $\mathcal{A}_n$, the dihedral slip reflections TP and PT, and the Euclidean sheet-plane counterexample.

Python

```
"""

Standalone Verification Suite for:

"Bicomplex Difference Calculus on Native Binary Rings: Boolean Mobius Conjugacy,

Recursive Center Nesting, and Reader-Relative Realization"

Author: Dean A. Kulik | QuHarmonics Research Group | ORCID: 0009-0003-3128-8828

Requires: Python 3.8+ and NumPy.

"""

import hashlib
```

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```
import numpy as np


# 1. Fundamental Word Involutions and Native Ring Construction

def J_int(x, n):

   return int(format(x, f'o{n}b')[::-1], 2)


def C_int(x, n):

   return x ^ ((1 << n) - 1)


def is_pal(x, n):

   return J_int(x, n) == x


def P_int(x, n):

   return C_int(x, n) if is_pal(x, n) else J_int(x, n)


def ring_list(n):

   reps = sorted(x for x in range(1 << n) if x < P_int(x, n))

   m = reps[-1]

   return reps[:-1] + [P_int(m, n), m] + [P_int(x, n) for x in reversed(reps[:-1])]


def ring_matrix(n):

   R = ring_list(n)

   N = 1 << n

   X = np.zeros((N, n), dtype=int)

   for k, val in enumerate(R):

      for j in range(n):

         X[k, j] = (val >> (n - 1 - j)) & 1
```

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```
    return X, R


# 2. Address-Axis and Word-Axis Matrix Operators over F_2
def mobius_matrix(n):
    N = 1 << n
    idx = np.arange(N)
    return ((idx[:, None] & idx[None, :]) == idx[None, :]).astype(int)


def shift_T(n):
    N = 1 << n
    T = np.zeros((N, N), dtype=int)
    for k in range(N):
        T[k, (k + 1) % N] = 1
    return T


def shift_S(n):
    N = 1 << n
    S = np.zeros((N, N), dtype=int)
    for k in range(N - 1):
        S[k, k + 1] = 1
    return S


def addr_flip(n):
    N = 1 << n
    Pi = np.zeros((N, N), dtype=int)
    for k in range(N):
        Pi[k, N - 1 - k] = 1
```

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```
    return Pi


def shape_matrix(n):
    G = np.zeros((n, n - 1), dtype=int)
    for i in range(n - 1):
        G[i, i] = 1
        G[i + 1, i] = 1
    return G


def lamp_matrix(n):
    G = np.zeros((n, n // 2), dtype=int)
    for k in range(n // 2):
        G[k, k] = 1
        G[n - 1 - k, k] = 1
    return G


# 3. Verify Theorems 3.2, 3.4, and 3.5 (Two-Axis Bicomplex & Mobius Conjugacy)
for n in range(2, 9):
    N = 1 << n
    X, R = ring_matrix(n)
    M = mobius_matrix(n)
    T = shift_T(n)
    S = shift_S(n)
    Pi_N = addr_flip(n)
    Pi_step = (T @ Pi_N) % 2
    J_n = np.eye(n, dtype=int)[::-1]
    J_nm1 = np.eye(n - 1, dtype=int)[::-1]
```

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```
    G_delta = shape_matrix(n)
    G_lamp = lamp_matrix(n)

    # Theorem 3.4: M^2 = I, M(I+T)M = S, M D = S A, and X = M S^T M D
    D_op = (np.eye(N, dtype=int) + T) % 2
    assert np.array_equal((M @ M) % 2, np.eye(N, dtype=int))
    assert np.array_equal((M @ D_op @ M) % 2, S)
    A = (M @ X) % 2
    D = (D_op @ X) % 2
    assert np.array_equal((M @ D) % 2, (S @ A) % 2)
    assert np.array_equal(X, (M @ S.T @ M @ D) % 2)

    # Theorem 3.5: Matrix Intertwining and Shape Mirror
    p = np.array([[1 if is_pal(val, n) else 0] for val in R], dtype=int)
    phi = (D_op @ p) % 2
    ones_n = np.ones((1, n), dtype=int)
    assert np.array_equal((Pi_N @ X) % 2, (X @ J_n + p @ ones_n) % 2)
    assert np.array_equal((Pi_step @ D) % 2, (D @ J_n + phi @ ones_n) % 2)
    Delta_X = (X @ G_delta) % 2
    Delta_D = (D @ G_delta) % 2
    assert np.array_equal((Pi_N @ Delta_X) % 2, (Delta_X @ J_nm1) % 2)
    assert np.array_equal((Pi_step @ Delta_D) % 2, (Delta_D @ J_nm1) % 2)
    assert np.array_equal((M @ phi) % 2, (S @ ((M @ p) % 2)) % 2)

print("PASS: Theorems 3.2, 3.4, 3.5 (Bicomplex, M D = S A, X = M S^T M D, Channel Intertwining) for n=2..8")

# 4. Verify Theorem 3.6, Theorem 4.1, and Theorem 4.3 across n=2..14
```

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```
def interval(a, k, n):
  return ((1 << k) - 1) << (n - a - k)


def alphabet_closed_form(n):
  A = {interval(a, k, n) for a in range(1, n) for k in range(1, n - a + 1)}
  if n % 2:
    A.discard(interval((n - 1) // 2, 1, n))
  A |= {interval(0, k, n) for k in list(range(1, n // 2 + 1)) + [n - 2, n]}
  if n % 2:
    A.add(int('0' * ((n - 3) // 2) + '101' + '0' * ((n - 3) // 2), 2))
  else:
    A |= {
      int('0' * (n // 2 - 2) + '101' + '0' * (n // 2 - 1), 2),
      int('0' * (n // 2 - 1) + '101' + '0' * (n // 2 - 2), 2),
    }
  counts = (
    (n + 1) ** 2 // 4 - 2,
    (n + 2) ** 2 // 4 - 4,
    3 * ((n - 3) // 2),
    n * n // 2 + 4 if n % 2 == 0 else (n * n + 3) // 2,
  )
  return A, counts


for n in range(2, 15):
  R = ring_list(n)
  N = len(R)
  h = N // 2
```

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```
    D = [R[k] ^ R[(k + 1) % N] for k in range(N)]

    # Theorem 3.6: Palindrome Isolation & Exact Flag Count
    pals = [k for k, x in enumerate(R) if is_pal(x, n)]
    assert all(pals[i + 1] - pals[i] > 1 for i in range(len(pals) - 1))
    flags = sum(1 for a in range(N - 1) if is_pal(R[a], n) != is_pal(R[a + 1], n))
    assert flags == (1 << ((n + 1) // 2 + 1)) - 2

    # Theorem 4.1: Center Nesting R_n \ Pal_n -> R_{n+2} (checked up to n=12 -> 14)
    if n <= 12:
        R2 = ring_list(n + 2)
        core = [x for x in R if not is_pal(x, n)]
        lo = len(R2) // 2 - len(core) // 2
        assert R2[lo:lo + len(core)] == [(1 << (n + 1)) | (x << 1) | 1 for x in core]

    # Theorem 4.3: Closed-Form Instruction Alphabet for n >= 6
    if n >= 6:
        rise, mirr = set(D[:h - 2]), set(D[h + 1:N - 1])
        A_pred, c_pred = alphabet_closed_form(n)
        assert set(D) == A_pred
        assert (len(rise), len(mirr), len(rise & mirr), len(set(D))) == c_pred

    # Proposition 6.3: TP and PT Fixed Points
    assert [a for a in range(N) if (N - a) % N == a] == [0, N // 2]
    assert [a for a in range(N) if (N - 2 - a) % N == a] == [N // 2 - 1, N - 1]

print("PASS: Theorems 3.6, 4.1, 4.3, and Proposition 6.3 verified for n=2..14")
```

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```
# 5. Verify Theorem 6.2 (Constructive Euclidean Sheet-Plane No-Go Counterexample)

N_atom = np.array([-1.458, 0.0, 0.0])

CA_atom = np.array([0.0, 0.0, 0.0])

C_atom = np.array([0.550, 1.420, 0.0])

CB_atom = np.array([-0.535, 0.780, -1.205])

u_tan = np.array([1.0, 0.0, 0.0])

n1 = np.cross(u_tan, np.array([0.0, 1.0, 0.0]))

n2 = np.cross(u_tan, np.array([0.0, 0.0, -1.0]))

chi = float(np.linalg.det(np.stack([N_atom - CA_atom, C_atom - CA_atom, CB_atom - CA_atom])))

assert abs(chi - 2.494784) < 1e-6 and (n1 @ (CB_atom - CA_atom)) * (n2 @ (CB_atom - CA_atom)) < 0

print(f"PASS: Theorem 6.2 Euclidean Counterexample (chi = {chi:.6f}, n1.CB = {n1 @ CB_atom:.3f}, n2.CB = {n2 @ CB_atom:.3f})")
```

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